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Review

Natural Language Processing in Healthcare: From Unstructured Data to Clinical Intelligence

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ABSTRACT

Natural Language Processing (NLP) has emerged as a critical methodology for leveraging unstructured clinical text in modern healthcare systems. The recent advancements in deep learning, domain-specific language models, and the advent of large language models (LLMs) have greatly improved clinical information extraction, decision support, and automated documentation. However, these developments have raised new methodological and ethical issues which reduce the generalizability of the applications to the clinic and the adoption of the practices.

This review provides a comprehensive and coherent examination of the development of NLP in healthcare, as well as clinical uses and implementation issues. We examine the performance of the various NLP paradigms in various clinical scenarios, and discuss common challenges like annotation bias, interpretability and safety issues related to generative models, and data heterogeneity. We do not give a descriptive inventory, but we stress analytical synthesis and clinical relevance.

The review wraps up with a discussion on future research trends such as multimodal language modelling, federated learning, explainable AI and governance of LLMs. This review addresses the gap between methodological developments and clinical application, providing a valuable reference for the research and implementation of NLP-driven healthcare solutions.

1. Introduction

Healthcare systems produce a large amount of unstructured textual information that are unstructured, such as clinical notes, radiology reports, discharge summaries, and narratives of patients (Papageorgiou et al., 2025). Even though these data include essential information to diagnose, risk-stratify and clinical decision-making, they cannot be systematically analyzed and reused on a large scale because of their unstructured nature (Worku, Arage and Kebede, 2025). The early solutions to extracting structured information through clinical text do exist based on traditional rule-based and statistical approaches but they tend to fail with linguistic variability, contextual ambiguity and

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domain-specific terminology (Rijcken et al., 2025).

The recent developments in natural language processing (NLP), especially domain-adapted deep learning and foundation models have significantly enhanced the capacity to extract clinical context, temporality, and semantic relationships in healthcare text (Velupillai et al., 2015). Consequently, NLP has become a vital part of a contemporary digital health infrastructure that, through automated documentation, clinical decision support, early risk identification, and population-level analysis, is applicable (Ahmad, Seo and Jeon, 2025). Nonetheless, the sheer scale of NLP processes has also presented major issues concerning the generalizability of models, their interpretability, bias, information privacy and their application to diverse healthcare environments (Jerfy, Selden and Balkrishnan, 2024, Wang et al., 2025).

Several reviews have been written on healthcare NLP, but the current literature remains incoherent. Numerous reviews are either task-oriented (e.g., named entity recognition or sentiment analysis) or highlight algorithmic progress without adequately discussing clinical application and limitations as well as inter-approach comparative performance. Other works give general summaries but extensively depend on descriptive lists of the applications, with minimal synthesis or critical assessment of the mechanisms and reasons of success or failure as NLP strategies are applied in practice.

Despite many reviews on healthcare NLP, the majority are technical model development reviews or a summary of application descriptions, and there is little discussion on the performance of these techniques in a real clinical environment. There is still no comprehensive comparative assessment of traditional NLP methods, transformer-based approaches, and new large language models, particularly regarding their clinical usability, generalizability, and safety. This review fills this gap by providing a clinically focused synthesis that connects methodological developments to practical implementation issues, offering insight into where NLP can deliver concrete healthcare benefits and the major challenges that must be addressed to enable its responsible and scalable implementation. To understand the current capabilities of healthcare NLP, it is important to examine how the field has evolved from early symbolic approaches to today's data-driven and foundation-model paradigms.

1.1. Review Methodology

This literature search was systematic and carried out in accordance with Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA). The electronic databases (PubMed, Scopus, and ScienceDirect) were searched to find relevant studies. The various

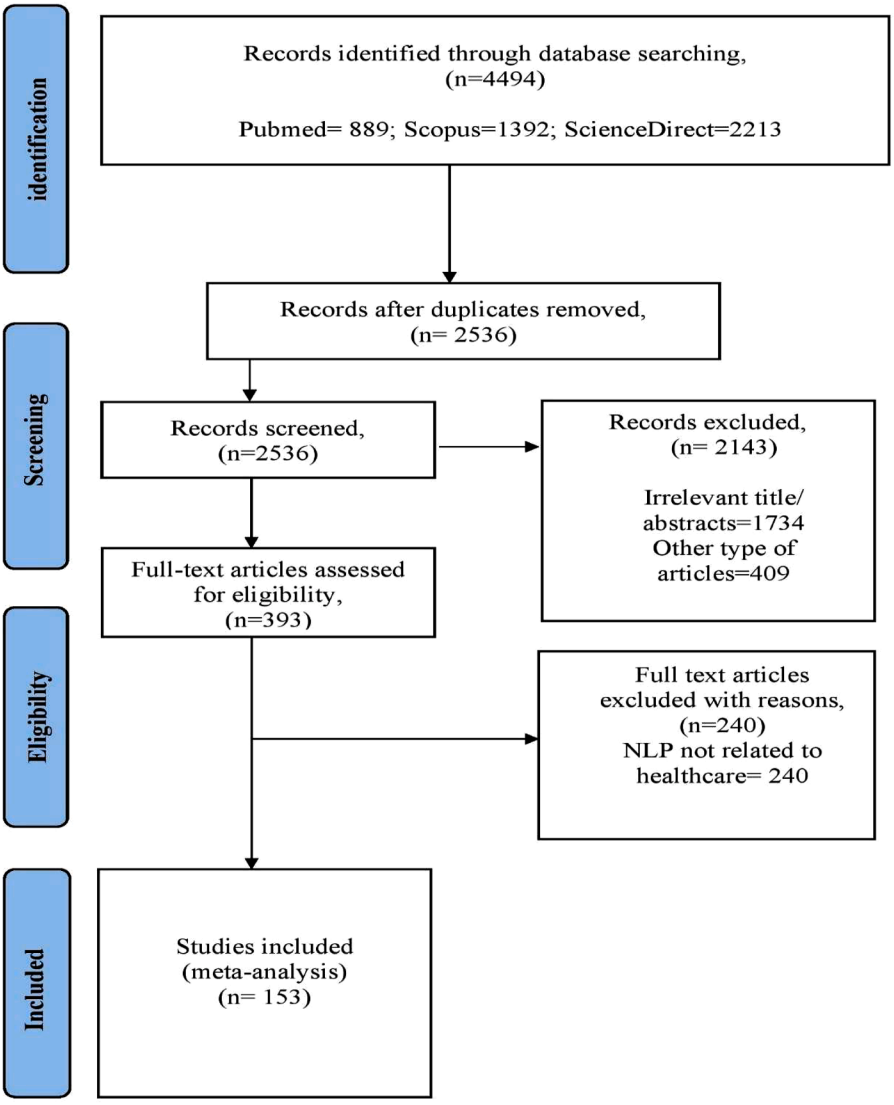


Figure 1. PRISMA Literature Selection Flowchart.

keywords used were natural language processing, artificial intelligence, clinical decision support systems, electronic health records, personalized medicine, and clinical intelligence. A total of 4494 records were retrieved from PubMed (889), Scopus (1392), and ScienceDirect (2213). The number of duplicate entries was removed and 2,536 unique entries were screened for titles and abstracts. At this screening step, 2,143 records were excluded, of which 1,734 had an irrelevant title or abstract, and 409 were other type of articles. The remaining 393 full-text articles were evaluated according to the pre-established inclusion/exclusion criteria. Among these, 240 full-text articles were removed due to the fact that they were not associated to the healthcare application of NLP. Overall, 153 studies were eligible and included in the qualitative synthesis and meta-analysis (Figure 1).

2. The Evolution of NLP

Historically, the development of NLP in healthcare has been primarily driven by the need to extract clinically meaningful information from unstructured medical texts, rather than by the broader advancements in the field of artificial intelligence (Delanerolle et al., 2025). In contrast to open-domain language applications, healthcare NLP must deal with domain-specific vocabulary, contextual ambiguity, time constraints, and strict regulatory requirements. This section describes the major shifts in methodology that have had a direct effect on healthcare NLP, paying particular attention to the advances that made the field suitable for practice (Wibawa and Kurniawan, 2024).

2.1. Early Theoretical Approaches (Symbolic or Rule-Based)

The initial NLP applications in health care were characterized by rule-of-thumb and statistical approaches that were developed to analyze structured clinical reports including discharge summaries, pathology reports and billing reports (Van Bulck, Reading Turchioe, Topaz and Song, 2025). These systems were based on manually constructed rules, dictionaries and domain ontologies to determine medical entities and relationships. Although useful with small scope projects like coding and basic information searching, rule-based systems were labor-intensive to maintain, and did not fare well with linguistic diversity amongst institutions and clinical disciplines (Aramaki, Wakamiya, Yada and Nakamura, 2022).

The emergence of statistical NLP techniques was a major change since they were the first models to learn patterns based on annotated clinical corpora (Khan et al., 2023). The methods like Hidden Markov Models, conditional random fields and early machine learning classifiers, which enhanced scalability and robustness especially in activities like named entity recognition and document classification (Kampatzis, Sidiropoulos, Diamantaras and Ougiaroglou, 2024). Nonetheless, these methods kept having issues in long-range context, in negation, in uncertainty and in time reasoning, which are key characteristics to clinical language interpretation (Klug et al., 2024).

2.2. Transition to Deep Learning and Domain Adaptation

The introduction of deep learning was a paradigm shift in healthcare NLP since it allowed models to learn more complicated semantic and syntactic patterns in a clinical text (Chen et al., 2025). Neural architectures enabled better dealing with the variability of clinical documentation and lessened the use of handcrafted features. Most importantly, this time also illuminated the constraints of directly using general-domain NLP models on healthcare data as text performance tended to drop as a result of vocabulary, syntax, and clinical reasoning bias (Rezaeianjouybari and Shang, 2020).

This drawback led to the creation of domain-adapted methods trained on medical literature and clinical literature (Ahmed et al., 2023). The domain adaptation enhanced the performance of a variety of clinical tasks, among which are information extraction, cohort identification, and clinical summarization. Notably, these developments saw healthcare NLP move away from experimental proof-of-concept systems in favor of systems that can support real-life clinical processes (Klug et al., 2024).

2.3. Modern NLP Paradigms in Healthcare

In the last 5 years, the clinical field of NLP has undergone a transformation with the rise of transformer-based systems and LLMs based on biomedical and clinical corpora (Kumar, 2024). In contrast to rule-based and machine-learning methods that use handcrafted features and domain-specific lexicons, modern NLP systems are pretrained on vast amounts of unstructured medical text and then fine-tuned or fine-tuned in-context for the specific task. In various clinical applications, such as named entity recognition (NER), relation extraction, clinical concept normalization, note summarization, question answering and disease risk prediction, this paradigm has been adapted to achieve state-of-the-art performance (Jadhav, 2023, O'Mahony et al., 2019).

The advancement of healthcare NLP is a journey from rule-based extraction systems to machine learning, deep learning, transformer-based models, foundation models, retrieval-augmented, multimodal AI, and the new generation of agentic models. The progression of methodological paradigms in health care NLP (models, clinical applications and workflow integration) is described in Table 1 (Garg and Gupta, 2024).

2.3.1. Domain-Specific Transformer Models

Researchers have created domain-specific transformer models, such as BioBERT and ClinBERT, to handle the specific vocabulary, abbreviations, and context complexity found in biomedical and clinical texts. These models are pre-trained on large-scale biomedical literature and clinical records, enabling them to better understand medical terminology and improve performance in healthcare NLP tasks such as named entity recognition, relation extraction, question answering, and clinical note analysis (Nerella et al., 2024). The major domain-specific transformer models developed for biomedical and clinical text processing are discussed in this section

2.3.1.1 BioBERT (Bidirectional Encoder Representations from Transformers for Biomedical Text Mining): BERT models are pretrained on biomedical literature, using PubMed abstracts and PubMed Central full-text articles. It is meant for improving biomedical text mining and information extraction tasks. BioBERT significantly outperforms them on the following three representative biomedical text mining tasks: biomedical named entity recognition (0.62% F1 score improvement), biomedical relation extraction (2.80% F1 score improvement) and biomedical question answering (12.24% MRR improvement). Later, it was demonstrated that BioBERT can consistently and significantly outperform generic transformers in terms of performance on medical term, abbreviatories and complex sentence structures common in biomedical literature (Lee et al., 2020).

2.3.1.2 ClinicalBERT: ClinicalBERT and its variant BioClinicalBERT are BERT-based models that are pre-trained using clinical text data like electronic health records , discharge summaries, progress notes, and clinical narratives. These models can enhance the understanding of clinical patterns and terminology in the domain-specific language of medical records, aiding in applications like disease prediction, clinical decision support, and healthcare information extraction. The ClinicalBERT model has set a new standard in clinical NLP tasks like phenotyping, ICD coding, adverse event detection, and clinical concept extraction (Li et al., 2023). Comparative studies confirm that ClinicalBioBERT excels in analyzing clinical notes, whereas BioBERT performs better on biomedical literature, highlighting the importance of domain alignment between pretraining corpus and target tasks (Vithanage, Yu, Wang and Deng, 2024).

2.3.1.3 PubMedBERT: PubMedBERT is a transformer-based language model that was trained from scratch on biomedical abstracts

Table 1
Evolution of NLP paradigms in healthcare with representative models and clinical relevance.

Methodological paradigm	Models	Clinical application	Workflow integration framework	References
Rule-based systems	cTAKES, MetaMap, MedLEE	Concept extraction, terminology normalization, EHR structuring	Low	(Lossio-Ventura, Sun, Boussard and Hernandez-Boussard, 2023, Reátegui and Ratté, 2018, Reátegui and Ratté, 2018)
Traditional Machine Learning	SVM, CRF, Random Forest	Disease classification, NER, phenotyping	Low-Moderate	(Ngugi, Akinyelu and Ezugwu, 2024, Idrissi Khaldi, Erraissi, Hain and Banane, 2025, Hatuwal, Shakya and Joshi, 2021)
Deep Learning Models	CNN, LSTM, BiLSTM-CRF	ADE detection, relation extraction, clinical event prediction	Moderate	(Shiri, Perumal, Mustapha and Mohamed, 2023, Yadav, Sharma and Yadav, 2024)
Transformer-Based NLP	BioBERT, ClinicalBERT, PubMedBERT	Information extraction, phenotyping, cohort identification, risk prediction	Moderate–High	(Madan et al., 2024, Shaji et al., 2025)
Clinical Foundation Models	GatorTron, BioGPT, Med-PaLM 2, MedAlpaca	Clinical summarization, question answering, decision support	High	(Singhal et al., 2025, Tofeeq, Naseer and Wali, 2025)
Retrieval-Augmented Generation (RAG)	MedRAG, EHR-RAG	Guideline retrieval, evidence-grounded QA, patient-record summarization	High	(Zhao, Liu, Yang and Miao, 2025, Neha, Bhati and Shukla, 2025, Abo El-Enen, Saad and Nazmy, 2025)
Multimodal Clinical AI	Med-Gemini, LLaVA-Med, Med-Flamingo	Multimodal diagnosis, image-text reasoning, treatment planning	Very High	(Yang et al., 2024)
Agentic Clinical AI Systems	Clinical Workflow Agents, Autonomous Medical Agents	Triage, care coordination, workflow automation	Transformative	(Zhao et al., 2026, Warrier and K, 2025)

collected from PubMed instead of fine-tuned by additional pretraining on the general domain model. It learns biomedical words and structures from biomedical literature, which will lead to deeper knowledge about biomedical terminology and concepts (Bobart et al., 2025). This model has been shown to be superior on many biomedical NLP tasks on benchmark datasets, outperforming many other BERT variants. PubMedBERT is also the backbone of the BLURB (Biomedical Language Understanding and Reasoning Benchmark) framework that provides a standard platform for assessing biomedical NLP models (Sarker et al., 2024).

In addition to these basic models, models specific for sub-domains and languages have been developed, for example MedBERT, pretrained on the N2C2, BioNLP and CRAFT datasets for biomedical named entity recognition, BlueBERT, modified for PubMed and biomedical abstracts, or region-specific models like medBERT.de for German clinical text (Rohanian et al., 2024). Knowledge distillation and continual learning also have spurred the development of lightweight variants of clinical transformers, such as DistilBERT, models capable of matching the performance of larger models with much lower computational resources and memory usage.

2.3.2. *Large Language Models for Clinical Text*

In recent years, the field of clinical NLP has been significantly reshaped by the emergence of LLMs. Interesting potential has been seen in general purpose LLMs like GPT-3, GPT-4 and beyond performing surprisingly well on medical tasks like the US Medical Licensing Examination, summarizing medical notes, and answering medical questions (Maity and Saikia, 2025). But these models have serious drawbacks in healthcare, such as hallucinations, limited domain specificity, privacy issues, and possible bias, that hinder direct application in the clinical workflow.

To overcome these problems, researchers have created LLMs for healthcare by applying domain-adaptive pretraining and instruction tuning to medical corpora. The models exhibit significantly higher accuracy on medical QA, information extraction and summarization compared to general purpose LLMs (Saha et al., 2025). Examples include instruction-tuned medical LLMs trained on clinical notes, medical textbooks, and guideline documents, which better align model outputs with clinical reasoning and terminology.

LLMs have shown remarkable medical capabilities including passing the US Medical Licensing Examination and generating clinically coherent text but general-purpose LLMs suffer from hallucinations, domain misalignment, and privacy concerns. To address these limitations, researchers developed instruction-tuned medical LLMs through domain-adaptive pretraining and instruction-following fine-tuning on medical corpora (Li et al., 2026). Instruction-tuned LLMs trained on a variety of medical instruction-following data (e.g., AlpaCare, which is LLaMA fine-tuned with 52k machine-generated medical instructions) are much better at generalizing and performing well in both medical and non-medical settings and tasks. Comparisons with other benchmarks have demonstrated that instruction-tuned medical LLMs perform up to 100 times better on automatic metrics and result in fewer errors that are more aligned with clinical reasoning, medical terminology, and safety requirements for tasks such as summarizing clinical notes, answering medical questions, and providing clinical decision support (Li et al., 2026).

To overcome these challenges, including hallucination and outdated knowledge, clinical LLMs have introduced a new architectural paradigm: Retrieval-Augmented Generation (RAG). At runtime, RAG can retrieve relevant information from electronic health records (EHRs), clinical guidelines, medical literature, and knowledge graphs, which allows LLMs to produce more accurate, context-aware, and up-to-date responses (Jiang et al., 2026). Research has demonstrated that RAG can improve performance to that of processing the entire chart while using a much smaller number of tokens. ReinRAG is an advanced text generation method that leverages reinforcement learning for enhanced clinical text generation, including reinforcement learning-based knowledge graph retrieval for discharge summary and patient instruction generation (Hao et al., 2025). Overall, RAG improves the reliability, efficiency, and clinical applicability of LLMs in healthcare settings.

The transformer architecture-based models and LLMs have also shown their efficiency as feature extractors in the EHR studies with high accuracy in antibiotic prescription extraction, risk prediction, and disease phenotyping (Van Veen et al., 2023, Kumar et al., 2025).

2.3.3. *Multimodal NLP: Integrating Text, Imaging, and Laboratory Data*

Clinical decision-making is a process that involves multiple data types such as clinical text (Physician Notes, Discharge Summary, and Pathology Reports), medical imaging (MRIs, CT scans, discharge images, and histopathology slides) and structured data (laboratory values and genomic data). Multimodal large language models (MLLMs) are models capable of handling multiple modalities, such as text and images, and provide holistic clinical information, allowing for multiple tasks to be executed with high accuracy, including diagnostic support, treatment

planning, and patient management (AlSaad et al., 2024).

Multimodal AI systems combine medical images with text reports, lab data and other clinical information to address the limitations of single-modality systems. These systems can automatically generate radiology reports based on the image findings with the corresponding regions of interest, help clinicians with more accurate diagnoses, analyze longitudinal information to monitor disease course and treatment response, and support visual question answering by integrating image interpretation with advanced clinical reasoning (Nam et al., 2025).

Recent developments have been multimodal federated learning architecture, which allows for data privacy while collaborating across institutions. These frameworks combine ResNet-50 – Feature Pyramid Network models with cross-modal attention mechanisms to link visual data from medical images with respective clinical text embeddings (such as lab reports, patient history, etc.) to facilitate cross-modal diagnosis without relinquishing data sovereignty and HIPAA/ GDPR compliance (Magalhães, 2026).

MLLMs in healthcare are built to handle multiple data modalities: text, images, and structured data, providing assistance with tasks such as diagnostics, treatment planning, patient management, and more, with exceptional accuracy. The models include explainable reasoning and safe generation, optimizing healthcare workflows (Krones et al., 2025). The next frontier includes real-time, multimodal diagnostic assistance where AI systems interpret imaging, text, and physiological signals simultaneously to support point-of-care decisions, as well as multimodal foundation models trained on billions of medical images–text pairs (AlSaad et al., 2024).

2.3.4. Task-Specific Applications and Clinical Integration

Modern transformer-based models and LLMs have been categorized into six key clinical NLP tasks: dialogue generation, question answering, summarization, text classification, sentiment analysis, and named entity recognition (Cho et al., 2024). For instance, clinical chatbots have been demonstrated to be possible with a model such as MedPIR; however, there are privacy and ethical issues that need to be addressed. While the models based on BioBERT can enhance the accuracy, they have limitations in dealing with complex medical terms. BioBERTSum and LLM-based summarizers can help clinicians to summarize medical texts, but longer sequences are still difficult to summarize (Bednarczyk et al., 2025). Better summarization of longer sequences remains challenging, yet BioBERTSum and LLM-based summarizers can help clinicians cover medical texts. Transformer models achieve state-of-the-art performance on phenotyping, ICD coding, and adverse event detection. However, the adoption of transformer-based models in healthcare is still limited, mainly because of the difficulty in interpreting the output, the risk of bias, privacy concerns, regulatory requirements, and the need to fit their integration process with the legacy EHR workflow.

As summarized in Table 1, healthcare NLP methodologies have evolved through successive paradigms, ranging from rule-based systems (e.g., cTAKES and MetaMap) to transformer-based models (e.g., BioBERT and ClinicalBERT), clinical foundation models, RAG architectures, multimodal AI systems, and autonomous clinical agents, each contributing to progressively higher levels of clinical integration and decision support.

2.4. Emergence of Foundation Models for Clinical Text

The arrangement of large-scale pretrained language models led to an even more advanced state of healthcare NLP, allowing the contextual representation of medical language at scale. The foundation models that were trained on biomedical and clinical corpora had significant improvements in contextual, temporal, and entity relationship perception in clinical stories (Luo, Deng, Yang and Luo, 2024). More complicated processes like longitudinal patient trajectory, automated reporting, and clinical decision support were supported using these models. Simultaneously, with the implementation of the foundation models, new issues unique to healthcare appeared, such as fear of bias amplification, interpretability, data leakage, and inter-institutional generalizability (Nerella et al., 2024). Consequently, recent studies have also been highlighting more evaluation in the clinical environment, explainability, and correspondence with clinical decision-making than with model performance alone.

2.5. Adoption Trends and Clinical Relevance

The development of healthcare NLP is characterized by a progressive move towards clinical value and integration instead of technical feasibility. Practical needs that have led to the adoption of EHRS include lightening of the documentation burden, better information retrieval within electronic health records, and early risk identification (Velupillai et al., 2018). As market expansion and additional investment are promising indicators of rising interest, meaningful adoption is still closely linked to observable clinical impact, interoperability, and reliability as opposed to the sophistication of an algorithm (Woldemariam and Jimma, 2023).

Figure 2 illustrates the distribution and temporal evolution of methodological approaches in healthcare NLP from 2008 to 2021. Machine learning (ML)-based methods represent the largest proportion of studies (48.1%), reflecting their broad applicability across clinical NLP tasks. Rule-based approaches (18.9%) remain consistently utilized throughout the study period, while deep learning (DL)-based methods (16.0%) show increasing adoption after 2016. The remaining studies (17.0%) comprise hybrid and alternative approaches, highlighting the diversity of methodological strategies (Sarker, 2021).

Rather than demonstrating complete paradigm replacement, the figure reveals a pattern of methodological evolution characterized by the coexistence of multiple analytical approaches. The range of methods within this methodological pluralism corresponds to the different needs of healthcare NLP, such as interpretability, data availability, computing resources, and regulatory conformity in method choice. On the clinical side, the use of the rule-based, ML-based and DL-based approaches will remain flexible to cover other healthcare applications. In general, Figure 2 shows that the development of NLP in healthcare is not a replacement of old approaches, but rather a synergy of several methodological approaches.

2.6. Global Trends and Market Forecast

Although market reports project rapid growth in NLP in healthcare and life sciences market worldwide, these projections should be interpreted with caution because they rely on vendor-reported revenues, proprietary estimation methods, and assumptions of sustained demand and implementation readiness. Figure 3 illustrates the estimated development of NLP in the healthcare and life sciences market in key areas worldwide between 2018 and 2030 in billions of USD. The stacked bar chart shows that the total market is growing steadily with a particularly quick growth after 2024. As a percentage, North America has the highest proportion across the period, which is also indicative of early adopters of NLP technologies, good digital health infrastructure, and heavy investment in healthcare AI. Europe is the next biggest contributor, having experienced consistent growth due to the increasing health informatics projects and regulatory encouragement of digital health developments. The Asia-Pacific area demonstrates the most rapid proportional growth, in particular, during the previous years, which suggests the rising usage of NLP technologies and the rapid digitalization and the rising healthcare demand. However, Latin America and the Middle East & Africa also have minor but also steadily growing shares in the market, pointing to the emergent adoption of the market. Generally, the number supports the international diffusion of NLP use in healthcare and in life sciences, with the uneven distribution of these numbers indicating the inequality in infrastructure, investments, and regulatory

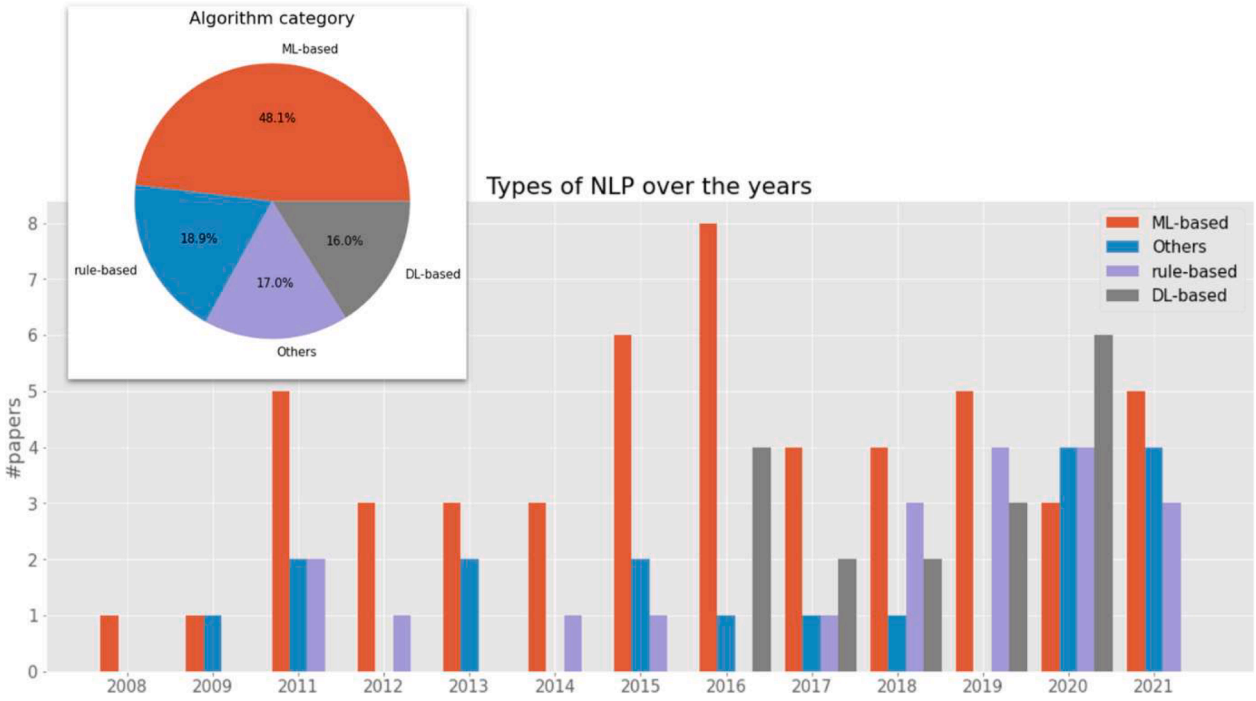


Figure 2. Trends in NLP algorithm usage over time. The bar chart shows the annual distribution of published NLP studies by algorithm category (machine learning–based, rule-based, deep learning–based, and other approaches) from 2008 to 2021, highlighting the early dominance of ML-based methods and the increasing adoption of deep learning in more recent years. The inset pie chart summarizes the overall proportion of each algorithm category across the full study period, with ML-based approaches accounting for the largest share, followed by rule-based, other hybrid methods, and deep learning approaches (Wu et al., 2022).

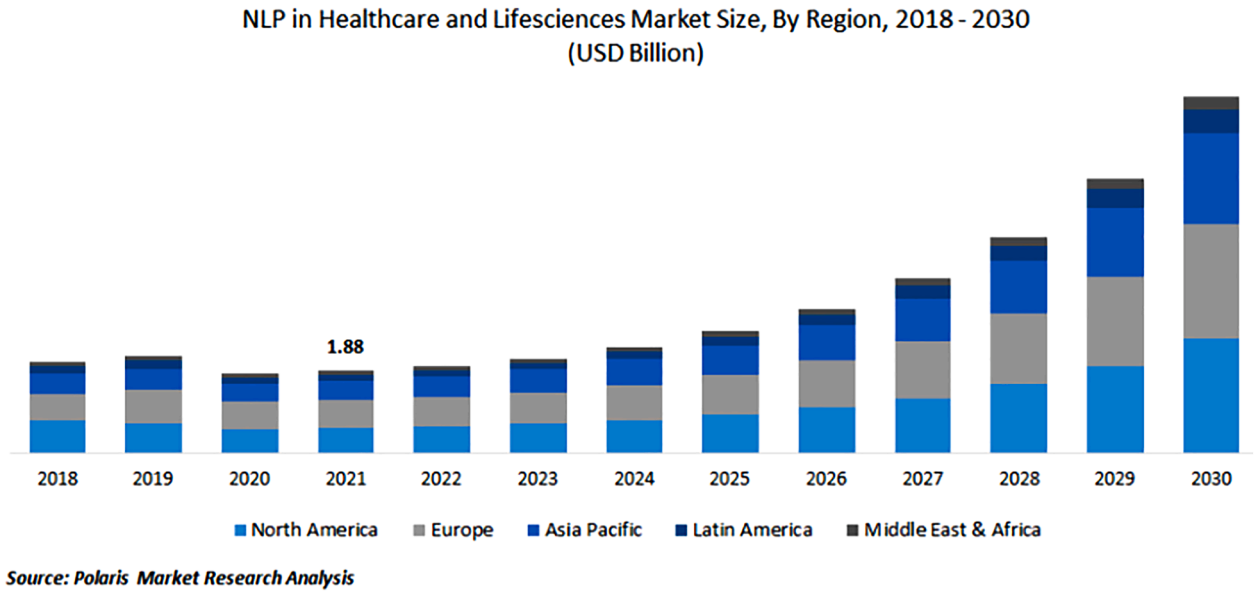


Figure 3. Projected market size of NLP in healthcare and life sciences by region from 2018 to 2030 (in billions of USD). The stacked bar chart highlights regional contributions, with North America and Europe leading the market share, followed by significant growth in the Asia Pacific region. The market shows a steady upward trend, particularly after 2024, reflecting increased adoption of NLP technologies across all regions. The market is projected to surpass $1.88 billion in 2021 and continue expanding rapidly through 2030 (Source: Polaris Market Research Analysis).

awareness.

Market analyses are helpful in understanding the emerging interest in NLP in healthcare but should be taken care when forecasting. Projections can be overly optimistic because they don't fully reflect regulatory hurdles, interoperability issues, provider readiness, coverage and payment concerns, and the varied implementation of digital health infrastructure among healthcare systems. Moreover, commercial and investor-driven reports may be biased to highlight innovative, growing markets, investment opportunities, and business successes, and downplay failures, scalability challenges, and lack of clinical success. Importantly, increasing commercial interest does not necessarily translate into demonstrated clinical impact, as many NLP applications still lack rigorous external validation, prospective clinical evaluation, and robust evidence of improvements in patient outcomes, clinician

efficiency, or workflow integration. Thus, along with projecting market growth, the characteristics of adoption trends must be examined by looking at the clinical evidence for safe, effective, and equitable adoption for clinical practice.

3. Components of NLP

The historical overview of NLP developments in history, we explain the fundamental functional elements that help such technologies to process and interpret clinical language in practice. The use of NLP in healthcare is contrasted with general-purpose NLP in that it must not only understand domain-specific language but must also be able to handle contextual uncertainty, and must also smoothly integrate with clinical processes (Rojas-Carabali et al., 2024). Instead of a rediscovery of the underlying theory of linguistics, this part is more about the functional aspects of NLP systems that are most applicable to clinical practices and how they can help in practical healthcare knowledge.

Figure 4 describes the end-to-end architecture of NLP, and it shows how raw textual input can be converted into useful and meaningful output using a series of closely related components and processes (Tucudean, Bucos, Dragulescu and Caleanu, 2024). On top, there is the illustration of the NLP workflow in general with the input text being processed through a successive series of pipelines of raw text processing, preprocessing, understanding, representation, generation and final output. This linear progression underscores the fact that NLP applications like translation, summarization, sentiment analysis, chatbots, question answering and information extraction uses the multiple interim phases instead of an algorithmic phase (Moon, Park, Yang and Jeong, 2022). The last Output Text can be in any form such as written answers, synthesized speech or optimized system responses. The main part of the diagram separates NLP into two fundamental subfields natural language understanding (NLU) and natural language generation (NLG), which are represented as complimentary yet dissimilar parts (Khurana, Koli, Khatter and Singh, 2023). NLU is concerned with the meaning of text and the role is to interpret the meaning of the user and linguistic structure. Other important processes in NLU are tokenization, part-of-speech tagging, named entity recognition, syntactic parsing, semantic analysis, and intent recognition (Karanikolas et al., 2025). Taken together, these tasks allow the system to know what is meant by the text, what objects the text refers to, and why it is written, which is necessary to interpret the text correctly in downstream applications. By contrast NLG, can be described as concerned with the generation of coherent, fluent and grammatically correct language out of internal representations (Chen, Zhang, Langrené and Zhu, 2025). The NLG process consists of content, document structuring, information, lexical selection, referring expression, and surface realization. These measures are taken to make sure the system does not just pass the right information but in a manner that is close to the way human beings communicate with each other (Astobiza, 2025). NLG has been especially critical in uses like report creation, conversational agents, as well as automated summarization. The division between NLU and NLG is a common base layer, comprising of text preprocessing, language models, machine learning and deep learning algorithms, knowledge bases, and organized information stores (Paaß and Giesselbach, 2023). This layer constitutes the computational core of NLP systems, which is a combination of both classical and modern methods like SVMs, random forests, n-grams, TF-IDF, WordNet, ontologies, and transformer-based models like BERT and GPT (Banou et al., 2025). These approaches facilitate comprehension and creation through giving illustrations, contextual concepts, and domain wisdom. In general, the figure highlights that NLP is not a unified methodology, but a multi-phase, modular framework where knowledge and creation are closely intertwined based on mutualized models and descriptions (Mansour, Aboelela, Talaat and Bustami, 2025). It also identifies the role of progress in machine learning and knowledge integration that allows NLP systems to transition away of raw text to meaningful and task-specific output in a broad application environment (Ho, Nguyen, Huy and Nguyen, 2025).

3.1. Clinical Language Understanding

Clinical language understanding is the process of NLP systems to understand the meaning of medical text in its clinical context. This involves being able to identify clinical entities (e.g., symptoms, diagnoses, and medications), overcome abbreviations, and deal with negation, uncertainty and time properly (Sharma and Kumar, 2025). In contrast to open-domain text, clinical narratives usually include short, fragmented sentences, and institution-specific norms, in which context-dependent interpretation is relevant. Appropriate language comprehension is at the basis of cohort identification, risk stratification, and decision support activities in healthcare settings (Steinkamp, Bala, Sharma and

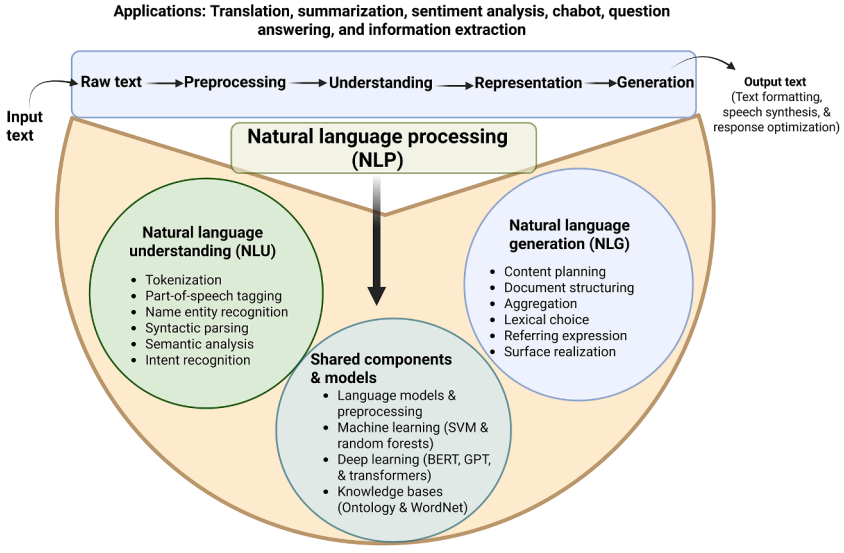


Figure 4. (a) Natural language understanding (NLU) focuses on interpreting and analyzing input text by using tasks like tokenization, parsing, semantic analysis, and intent recognition. (b) Core NLP techniques provide backbone methods (ML/DL models, language models, knowledge bases, text preprocessing) that support both understanding and generation. (c) Natural language generation (NLG) deals with producing human-like output by planning, structuring, choosing words, and forming grammatically correct text (Khurana, Koli, Khatter and Singh, 2023).

Kantrowitz, 2020). Errors made during this phase may be transferred to the downstream level causing wrongful categorization of patient status or with improper clinical advice. Thus, clinical NLP systems are usually considered not just in terms of extraction accuracy but also in terms of their inter-institutional and trans-specialty strengths (Velupillai et al., 2018).

3.2. Knowledge Structuring and Representation

The key objective of healthcare NLP is to transform free-text clinical narratives into structured and interpretable representations. It includes mapping extracted data to canonical clinical concepts and connecting entities via clinically significant relationships, e.g. drug event or symptom diagnosis relationships. Structured representations allow connectivity with the electronic health records, clinical decision support systems and population-level analytics (Gao, Mahajan, Uzuner and Yetisgen, 2024). Quality knowledge representation helps to longitudinally analyze patient trajectories as well as interoperate among health information systems. Significantly, this step makes healthcare NLP and generic text analytics very different in prioritizing interpretability and clinical validity rather than statistical associations only (Shaik et al., 2024).

3.3. Integration into Clinical Workflows

NLP systems need to be integrated into the current healthcare processes to bring actual clinical value. This incorporates connectivity to electronic health records, reporting and clinical dashboards. NLP systems that are workflow aware put more emphasis on the delivery of information promptly, reduce the amount of cognitive load on the clinicians, and match the output to clinical decision points (Goldhaber et al., 2024). Practically, the integration makes possible such applications as automated documentation, generation of alerts, and patient history summarization. Systems that cannot be integrated to workflow challenges, even when the technology is doing well, tend to be hindered by barriers to adoption as a result of usability and trust issues (Ye, Woods, Jordan and Starren, 2024).

3.4. Human Oversight and Clinical Validation

The medical NLP field works in a high-stakes setting where a mistake could lead to the direct impact on a patient. Therefore, the human control is a vital part of NLP implementation. Clinical validation, such as expert check and continuous monitoring is critical in enforcing reliability, fairness, and safety (Sharma, 2025). Most successful healthcare NLP systems tend to be used as decision-support systems, not due to complete autonomy, but as a complement to clinical judgement. This paradigm of collaboration emphasizes the relevance of transparency, explainability and feedback mechanisms in practical applications (Moreno-Sánchez, Del Ser, van Gils and Hernesniemi, 2026).

Figure 5 gives a conceptual process map that NLP converts raw healthcare text into clinically actionable information in a series of related processes. It starts with advanced text input, containing heterogeneous sources, including electronic health records, clinical notes, and medical documents, and patient reports (Chung, Song and Son, 2024). These inputs are not very structured and need to be processed in a specialized manner to make them clinically useful. The second phase, which is DL comprehension, entails analysis of the given text by the NLP system based on semantic analysis, context retrieval, and medical ontologies. This step aims at making sense of clinical meaning, and not superficial text, which allows the system to identify pertinent concepts, relationships and intent within complicated medical texts. This step is important since any error made at this point will be transmitted in the later processing stages (Fahim, Hasani, Kabba and Ragab, 2025). The third step is an intelligent feature extraction in which clinically significant data is extracted and converted to structured data. This involves the extraction of key medical terms and events as well as attributes in free text and their mapping onto standardized representations (Rani and Jain, 2023). NLP systems are useful in processing unstructured clinical data by transforming narrative information into formatted features, which can be used in the subsequent analysis. In the fourth step, more sophisticated algorithm processing, machine learning and predictive modeling is applied to the features that have been extracted to carry out more high-level analytics. These models are useful in the prediction of diseases, risk stratification of patients and clinical classification (Chen et al., 2020). This step is the shift between the information extraction and decision-based analysis, with NLP findings having a direct impact on clinical reasoning and risk assessment. The last step provides a full-fledged output, converting model output into clinical insight. These products can be individualized, provide tailored care advice, clinical advice, and action summaries and can be inserted into healthcare procedures. The highlighted uses of the end-to-end NLP pipeline at the bottom of the figure include disease prediction, personalized treatment, clinical documentation, and patient risk assessment, which are used to sum up the practical impact of the end-to-end NLP pipeline (Kreimeyer et al., 2017). In general, the figure highlights that NLP is a multi-stage enabling framework in healthcare that can gradually convert unstructured text into structured knowledge and eventually decision-support outputs that can promote patient care and clinical efficiency.

4. Clinical Applications of NLP in Healthcare

The various components of these foundational elements of NLP are

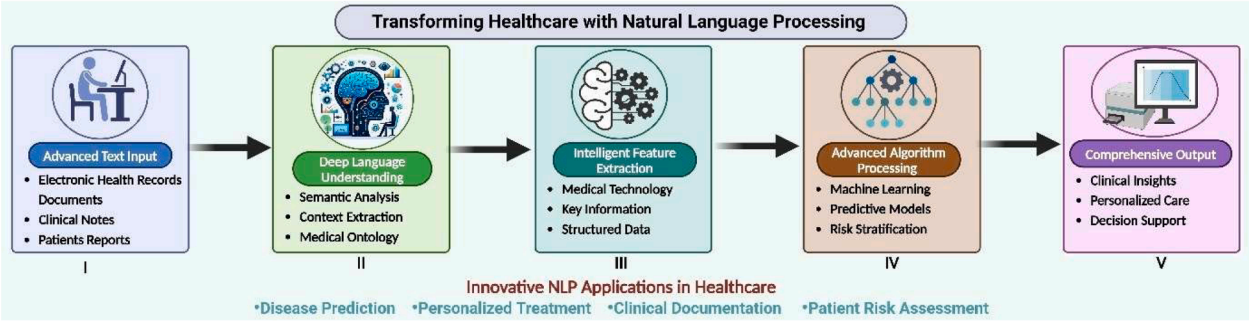


Figure 5. (I) Advanced Text Input: Unstructured data from electronic health records, clinical notes, and patient reports are collected as the foundation for NLP processing. **(II) Deep Language Understanding:** Semantic analysis, context extraction, and medical ontologies enable accurate interpretation of complex medical language. **(III) Intelligent Feature Extraction:** Key clinical terms, structured data, and relevant information are extracted to convert free text into usable features. **(IV) Advanced Algorithm Processing:** ML and predictive models analyze extracted features to generate risk assessments and clinical predictions. **(V) Comprehensive Output:** The system delivers clinical insights, personalized care strategies, and decision-support tools for improved patient outcomes (Clay et al., 2025).

discussed in relation to their application in healthcare. Instead of being a more general data-processing tool, NLP has demonstrated itself to be most useful in focused clinical applications, where it has been used to document, make decisions, and enhance workflow efficiency. Based on this, this section concentrates on practical clinical application, impact, and implementation limitations not on technical model architecture (Badalotti et al., 2024).

Figure 6 gives a task-based perspective of how various NLP methods are applicable to various sources of healthcare texts to create certain clinical and operational results. On top, it depicts the general workflow where the advanced text input is analyzed using NLP techniques to generate meaningful output and results. The core NLP layer is broken down into three fundamental capabilities, namely deep language understanding, intelligent feature extraction, and advanced algorithmic processing capabilities that are combined to convert raw text into actionable insights. The bottom panels show real applications of data sources, NLP methods, and outcomes, which indicate the value addition of NLP in various healthcare conditions. Optical character recognition (OCR) is used to process unstructured text and scanned data to allow digitization of the records so that legacy and handwritten documents can be used in electronic systems. The analysis of social media data and patient comments is used to obtain patient opinion, emotions, and the perception of the population to aid in patient experience analysis and monitoring on a population scale. Text classification and computer-assisted coding of medical records enhance the accuracy of the diagnostic process, treatment planning, and save on time and manual labor. The named entity recognition is used to analyze clinical data and patient data to identify key clinical concepts like diseases, drugs, and procedures and match patients to clinical trials. Additional analysis of clinical documents is performed by clinical relation extraction models that reveal correlations among medical objects (e.g., drug-disease, symptom-outcome, etc.), which in turn support focused treatment and clinical records. Lastly, with large-scale sources of text, such as social media, medical records, and research articles, there is an integration into advanced predictive modelling, which allows predicting the disease outbreak and forecasting the way a patient will respond to treatment. In general, the number highlights that NLP in healthcare does not represent a single technique but a multifamily of specialized approaches to various data sources, each of which provides a different and clinically significant result. The diagram explicitly connects the types of input data, NLP activities, and outcomes, which demonstrates the practical importance of NLP in enhancing efficiency, decision-making, and patient care throughout the healthcare continuum (Alafari, Driss and Cherif, 2025).

4.1. Clinical Documentation and Information Management

Clinical documentation support is one of the most developed and popular NLP applications to healthcare. The applications of NLP systems regularly include deriving structured information out of free-text notes, automating coding and billing systems, and producing summarized

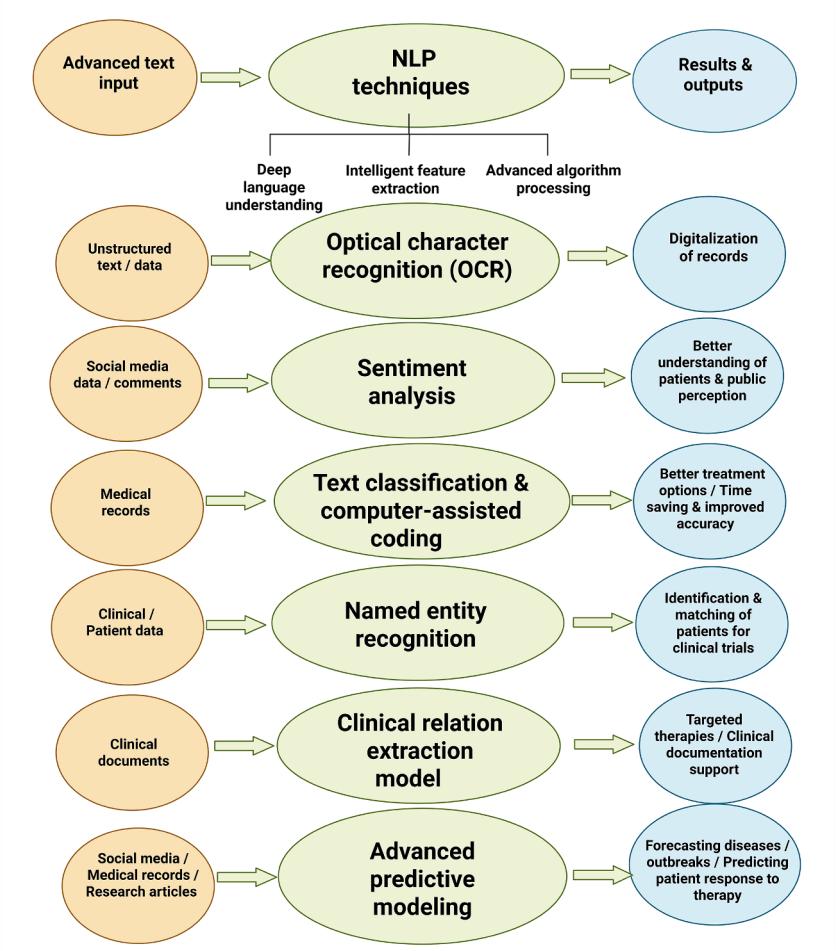


Figure 6. NLP in healthcare enables efficient processing of medical text by digitizing records through OCR, analyzing patient feedback with sentiment analysis, classifying clinical tests for accurate decision-making, and extracting key patient details using NER. Further, relation extraction uncovers drug–disease and treatment links, while predictive modeling helps forecast diseases and patient outcomes, collectively advancing personalized and data-driven healthcare (Wibawa and Kurniawan, 2024).

clinical notes. These applications will help reduce a significant cause of clinician burden and documentation-related burnout, namely, manual data entry and redundancy (Perkins, Muste, Alam and Singh, 2024). NLP-based documentation tools have been demonstrated to enhance complete and consistent data entry in electronic health records in clinical practice. But their success largely depends on the correct contextual interpretation, especially when it comes to dealing with negation, uncertainty and time references. Mistakes in these domains might result in poor documentation and future clinical risk, which is why special care is required concerning the validation and human control (Maguluri, 2025).

Figure 7 demonstrates that an electronic health records (EHR) system can be considered a central node that can combine different clinical data to predict diseases, provide individual treatment and encompass clinical records, which ultimately can promote health care. The EHR system at the center is a source of information that consolidates data of various sources and assists in interrelating key clinical functions. The disease predictor module integrates the inputs of vital signs, personal statistics, billing and risk assessment indicators in order to assist in the early detection of health risks and support custom-tailored medicine. These predictive features are relevant to the process of enhancing the efficiency of insurance and lowering the expenses, as well as proactive planning of care. The personalized treatment aspect makes use of demographic factors of the patient, history of medication, and diagnosis information so that the treatment plans are more specific and clinical. Meanwhile, the element of clinical documentation combines radiographic images, laboratory tests, and immunization and helps to maintain accurate and comprehensive records of patients and enables continuity of care. These essential functions are surrounded by the figure that focuses on the expanded impacts of successful EHR integration. These are better coordination of care, increased patient safety, simplified access to patient information, enhanced support of clinical decision-making, the capability to track patient data over time and improved and complete documentation. In summary, the figure underlines that centralization of heterogeneous clinical data and assisting predictive analytics, personalization, and documentation, EHR systems are essential to improving efficiency, quality and outcomes in contemporary healthcare.

Table 2 summarizes a selection of typical NLP studies in healthcare, in the form of a study type, AI models, and datasets used, with an example of the methodological diversity of clinical NLP tasks. In tasks related to segmentation, Support Vector Machines and Hidden Markov Models, which are based on traditional probabilistic and margin-based models, have been used with hand-crafted or labelled clinical data, as a part of early attempts to model clinical text. Medical coding research is showing a trend towards deep learning, with convolutional neural networks and transformer options utilized to access large-scale intensive care data to automate the process of code assignment. Under de-identification, sequence modelling models like conditional random fields and long short-term memory networks have been applied to detect and eliminate secured health data in electronic health records, frequently based on clinical NLP curated datasets. Patient embedding studies use deep learning and transformer-based networks to learn the longitudinal patient representation of large epidemiological or institutional data so as to perform subsequent downstream tasks, including risk prediction and cohort analysis. Hybrid architectures where recurrent neural networks are trained with structured prediction layers are typically utilized to extract the clinical entities and their relation from an annotated dataset to extract the information. Recurrent and transformer-based models dominate EHR text generation and summarization tasks and are used to generate coherent clinical text or summaries of structured or semi-structured input using both publicly available and proprietary data. Lastly, translation studies in medical language use multilingual transformer models already trained and specialized on clinical terminology which allows cross-lingual communication and exchange of information in healthcare settings. On the

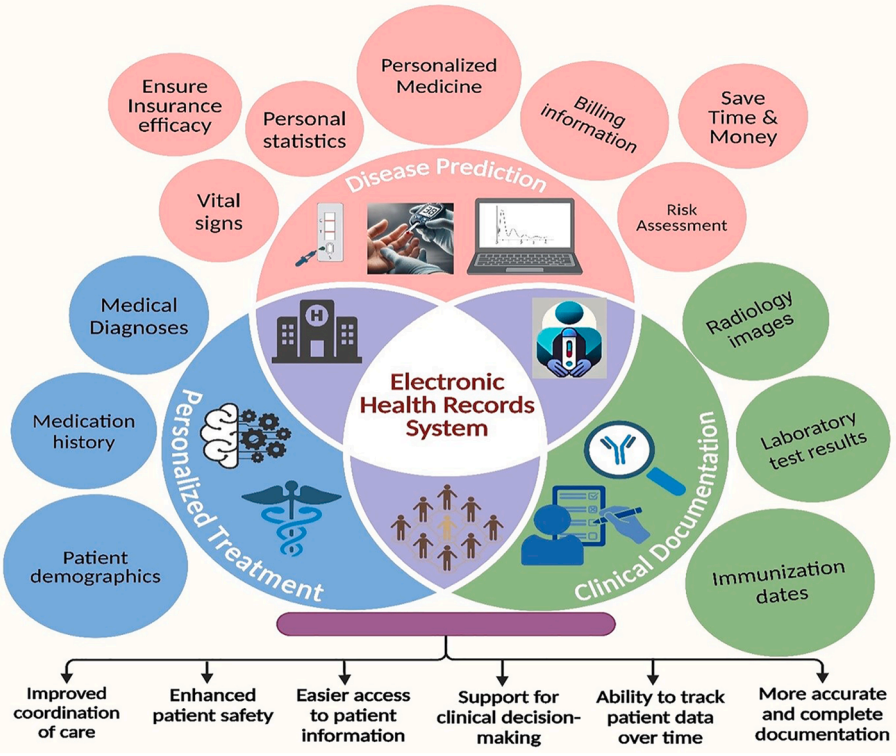


Figure 7. NLP in healthcare enhances data utilization by converting speech into structured records, extracting critical information from clinical texts, and classifying reports for better diagnosis and treatment planning. It supports information retrieval, sentiment analysis of patient experiences, and question-answering for quick access to medical knowledge, thereby improving decision-making and patient care efficiency (Ramakrishnaiah et al., 2025).

Table 2
Implementation of NLP in unstructured EHR.

Sr No.	Type of study	AI model	Dataset	Reference
1	Segmentation	Support Vector Machine (SVM)	Hand-crafted training dataset	(Khalate, Gite,
2		Hidden Markov Model (HMM)	Labelled clinical dataset	Pradhan and Lee, 2024)
3	Medical coding	Convolutional Neural Network (CNN)	MIMIC-III	(Edin et al., 2023)
4		Bidirectional Encoder Representations from Transformers (BERT)	MIMIC-III	
5	De-identification	Conditional Random Field (CRF), Long Short-Term Memory (LSTM)	EHR and MedNLP-1	(Aloqaily et al., 2025)
6	Patient embeddings	Deep Learning	Rochester Epidemiology Project data	(Xian et al., 2024)
7		Bidirectional Encoder Representations from Transformers (BERT)	Private data	
8	Information extraction	Long Short-Term Memory (LSTM)-Conditional Random Field (CRF)	i2b2	(Ji, Chen and Jiang, 2020)
9	EHR generation	Long Short-Term Memory (LSTM)	MedText	(Li et al., 2023)
10		Transformer	MIMIC-III	
11		Auxiliary Signal-Guided Knowledge Encoder-Decoder (ASGK)	CX-CHR and COVID-19-ΔΔCT report	
12	Summarization	Long Short-Term Memory (LSTM)	Own	(Luo et al., 2020)
13	Medical language translation	Pre-trained Multilingual BERT – Machine Translation model	MedLane	(Luo et al., 2020)

whole, the table demonstrates the evolution of conventional machine learning algorithms to new and improved deep learning and transformer-based algorithms in a wide scope of healthcare NLP applications.

4.2. Clinical Decision Support and Risk Stratification

In present-day clinical decision support systems, NLP has become a key component as it facilitates the retrieval of pertinent patient data in unstructured text and its association with structured clinical data (Berge et al., 2023). The functions of such systems include the recognition of high-risk patients, the detection of possible adverse outcomes, and the application of the appropriate clinical guidelines at the point of care (Balogun, Nashid and Khan Papia, 2025). Instead of substituting clinical judgement, NLP-enhanced decision support instruments act as assistant technology, which puts clinically relevant data on the priority list. They have the greatest influence in environments where there is an overload of information, like in intensive care, as well as in emergency departments (Musik, Sasin-Kurowska and Panczyk, 2025). However, in different institutions, the performance might differ because of the dissimilarity in documentation practice, which highlights the significance of local assessment and modification (van de Burgt et al., 2023).

4.3. Early Disease Detection and Phenotyping

Clinical narrative patterns that are not necessarily reflected in structured fields have been used extensively in NLP to detect diseases at their early stages. It has found use in the detection of disease progression, in the identification of diseases not known, and in the phenotyping of heterogeneous or complex diseases (Faviez et al., 2025). The apps are especially useful in conditions whose clinical manifestations are subtle or dynamic like mental disorders, rare diseases, and chronic illness at an early stage (Rani et al., 2025). NLP-based phenotyping reported encouraging outcomes, but its validity is based on the quality of data and its context over time. Disparate records and sporadic records are also a major obstacle to large-scale implementation (Duque et al., 2025).

4.4. Patient Experience and Communication Analysis

Outside of clinician-facing applications, NLP finds more applications in the analysis of patient-generated text, such as surveys, messages, and remote monitoring inputs. The analysis can help reveal patient experience, adherence, and perceived quality of care, thus resulting in more responsive and patient-centred healthcare delivery (Khanbhai et al., 2022). These tools aid in remote monitoring and subsequent care, especially in chronic diseases, in the clinical context. Nonetheless, the issue of interpretability and bias is still present because the language of the patients can be extremely diverse due to their cultural, socioeconomic, and linguistic backgrounds. Special attention to modeling should be paid to prevent wrong interpretation or unfair results (Jim et al., 2024).

4.5. Population Health and Research Enablement

At the population level, NLP enables the analysis of clinical text on a large scale, used in surveillance, cohort identification, and research of outcomes. NLP allows epidemiological studies to be wider and helps health systems to plan by making available information that can be found within free-text records (Stewart, Chaturvedi and Roberts, 2024). The latter applications will be the most efficient when used together with stringent data regulation frameworks and clear reporting of model constraints. Although NLP is able to speed up research and population health surveillance, results must be put into perspective with conventional statistical and clinical experience (Fallatah and Adekola, 2024).

5. NLP as a Permitting Layer in Digital Health Technologies

NLP is not a complete solution to healthcare technologies, but a cross-cutting interpretative layer, which converts human language into practical clinical knowledge (Dai et al., 2025). NLP is not just repeating the features of related technologies, but instead it is about how NLP can augment or make a difference in terms of the practical use of these systems (Clay et al., 2025). In addition to individual clinical activities, NLP is becoming a facilitating layer that links a variety of digital health technologies, which can further reach integrated healthcare communities.

5.1. Internet of Medical Things (IoMT)

IoMT devices are the main sources of physiological signals, and clinical interpretation may rely on the textual information provided along with the physiological results, patient-reported outcomes, and clinician annotations to a large extent (Alsaeed and Nadeem, 2025). NLP allows the IoMT platforms to combine the inputs of these narratives and sensor data by deriving structured clinical meaning in free text and speech (Khan et al., 2025). NLP is used to promote more contextualized monitoring through this integration, where the systems can differentiate between clinically relevant variability and benign variability (Alturki et al., 2024). In the absence of NLP, IoMT systems can only work with

numbers and thresholds, but with NLP, it is possible to integrate patient context and subjective symptoms as well as longitudinal clinical histories into the decision-making process (Osama et al., 2023).

5.2. Telemedicine and Virtual Care

Telemedicine encounters massive amounts of chatter data that are infrequently materialized upon. NLP allows machine translation, summarization, and analysis of the interaction between patients and clinicians, and converts virtual appointments into structured clinical data (Cunha Reis, 2025). NLP is also used in remote interactions and has been applied in addition to documentation to facilitate automated triage, clinically salient concerns, and emotional or cognitive indicators. It improves continuity of care and lowers documentation load, which makes NLP an enabler of scalable and high-quality virtual healthcare and not an auxiliary provision (Haleem, Javaid, Singh and Suman, 2021).

5.3. Wearable Devices and Real-Time Analytics

Smartwatches, biosensors, and fitness bands are modern examples of wearables collecting both physiological and behavioral data. NLP can process free-text data (user voice commands, e.g., pain diary, mood entries, etc,) and match it to sensor data. Contextual health monitoring is also improved by using NLP as a combination of textual symptom descriptions and EHR cardiovascular-based measurements in real-time, like heart rate or sleep quality. The combined knowledge will be able to show alerts, enhance follow-up of therapies, and resource longitudinal research on persistent diseases (Vijayan et al., 2021, Paganelli et al., 2022). The data captured is sent to edge devices or cloud platforms for real-time analytics and computation. Using real-time analytics, patients can be recognized on time, and medical assistance can be provided. This device has applications in sports science, disaster rescue, disability assistance, location tracking, and many more. The data stored on the cloud platform through wearable devices can be processed using various Big Data Analytics algorithms, and makes all patient information available to different specialists in the hospital.

5.4. Blockchain in Health Information Management

Blockchain would provide safe and unalterable storage of health-related data, whereas NLP would improve the extraction and harmonization of knowledge within the distributed sources. Blockchain-dependent health records can be annotated automatically by NLP, which allows for the easy acquisition and summary of clinically relevant material. Furthermore, blockchain smart contracts powered by NLP can provide enforcement of the consent protocol and a record of action, provided that the linguistic input complies with the required data in sensitive functions like clinical trials or sharing patients' information (Ghafar and El-Masry, 2021, Han, Zhang and Vermund, 2022). Blockchain supports a distribution network structure as opposed to a traditional centralized network structure for the completion, storage, and verification of transactions. Security, privacy, trust, and decentralization can be enhanced in a healthcare supply chain operation by implementing blockchain technology (Kumar et al., 2026). Sensitive information like patient details, the genesis of disease-related knowledge, the initial stages of drug research and development, the rate of discovery of drugs, the challenges experienced during drug research, and the transportation and storage conditions of medicine and temperature-sensitive products can all be shielded during the bidirectional information exchange between stakeholders using blockchain encryption and decryption techniques (Kumar et al., 2026).

5.5. AI and Machine Learning Synergies

NLP is an important intermediary to unstructured clinical language and machine learning models, which take structured inputs (Eguia et al., 2024). Through NLP, predictive modelling, risk stratification, and decision support may be achieved in various AI-driven healthcare applications by extracting clinically meaningful features in free text (Bazoge, Morin, Daille and Gourraud, 2023). Notably, NLP defines what is translated in the input of the clinical language into the models, which has a direct effect on the resultant downstream performance, bias, and interpretability. In this regard, NLP cannot be merely an upstream preprocessing step and is the determinant of AI system reliability in a healthcare setting (Fahim, Hasani, Kabba and Ragab, 2025).

5.6. Precision Medicine and Genomics

Less traditional, NLP and 3D printing combination is receiving growing interest in patient-specific modeling and surgical planning. The clinical narrative can be interpreted with NLP to reach anatomical description or surgical needs, which can be converted into design parameters of the 3D-printed models. As an example, the clinical notes about tumor size or implant specifications can be transformed into 3D prototypes, which will result in fewer pre-operative planning errors and increase the precision of the medical outcomes (Zhao and Yam, 2024, Sun, 2020). Additionally, 3D printing is now gaining ground in the development of personalized medicine and the printing of artificial tissue, moving beyond simple production. For example, the University of Wollongong in Australia used 3D printing to develop bones and muscles. Moreover, a concept for 3D bio-printing entails a complete reimagination of the core technology. This innovative concept involves full-color embedding of living cells, resulting in printed objects with active biological properties (Verma et al., 2025).

6. Methodological Landscape of NLP in Healthcare: Capabilities and Limitations

The development of healthcare NLP has gone through a series of methodological paradigms, each with its own advantages and shortcomings of use with clinical text (Lederman, Lederman and Verspoor, 2022). Although these applications indicate the scale of NLP implementation, to evaluate their performance, it is essential to take a closer look at the methodological paradigms, which govern the performance and constraints (Nazi and Peng, 2024).

6.1. Rule-Based and Hybrid Approaches

Transparency and interpretability make rule-based and hybrid NLP systems applicable in the healthcare industry. These methods are based on established linguistic principles, which are clinical lexicons and ontologies to pick out textual information (Alnattah et al., 2025). Their main advantage is that they are predictable and readily validated, especially in regulated clinical conditions. Rule-based systems are, however, not scalable across institutions and specialties since differences in clinical language will have to be maintained manually (Berge et al., 2023). The use is typically characterized by poor performance when used with new data sets or changing documentation habits. Consequently, these systems are best suited to small scope endeavors or elements of hybrid pipelines as opposed to being a stand-alone solution (Withers et al., 2025).

6.2. Traditional Machine Learning Models

Greater flexibility was provided in classical machine learning methods, which learn patterns in annotated clinical text. In activities like document classification and named entity recognition, these models tend to perform better than pure rule-based systems, especially when there is enough labelled data (Wu et al., 2017). Although the classical machine learning schemes have been scaled better, they are still ineffective regarding long-range dependencies, contextual peculiarities, as

well as temporality in clinical narratives; they are also sensitive to feature engineering decisions (Seow et al., 2025). Having an over-dependence on task-specific features also restricts their ability to be generalized across institutions, and retraining or recalibration is required to be deployed outside of the source of the data (Hussain et al., 2021).

6.3. Deep Learning and Contextual Models

Deep learning models represented a significant step forward in healthcare NLP, as they reduced the reliance on hand-designed features and provided richer semantic representations. Contextual models have proved to be more accurate with information extraction, clinical summarization, and phenotyping, even in large and well-curated datasets (Krones et al., 2025). These gains, however, have significant trade-offs. The typical features of deep learning models are large amounts of annotated corpora, high computer requirements, and overfitting, which can be easily prevented by careful tuning. In addition, the improvement of performance levels in benchmark datasets frequently reduces in clinical practice with documentation heterogeneity and domain shift (Malashin et al., 2024). Such restrictions make the process of generalizing research settings to clinical practice more difficult as discussed in Table-3.

Recent transformer-based models consistently outperform traditional machine-learning approaches. ClinicalBERT and BioBERT achieve F1 scores approaching 90% for clinical named entity recognition tasks, whereas classical CRF and SVM systems typically report F1 scores between 75–85%. Large language models such as Med-PaLM 2 and GPT-4 further improve reasoning-intensive tasks, although concerns regarding hallucination and reproducibility remain.

Table 4 shows a comparative analysis of top clinical and biomedical NLP models in the healthcare domain. Traditional transformer-based models like BioBERT, ClinicalBERT, and BioClinicalBERT performed well in NER and de-identification tasks with F1-scores ranging from 87% to 92%, showing their ability to extract clinical information from medical texts. Overall, PubMedBERT achieved good biomedical NLP performance, and its performance on clinical classification tasks was especially high. However, specialized models such as CheXbert achieved high precision in reporting radiology reports, RoBERTa-MIMIC and DistilBERT-Clinical models achieved good results in phenotyping, NER, and PHI detection. Recently, large language models like Med-PaLM and GPT-4 Clinical have been shown to be effective in QA tasks and clinical reasoning across medicine, with GPT-4 Clinical achieving the highest accuracy (86–90%). In general, the findings suggest a shift from task-specific NLP models to more sophisticated LLMs with the ability to understand and reason more broadly about clinical situations.

Table 3
Comparative analysis of major NLP paradigms in healthcare: Strengths, limitations, and clinical suitability.

Paradigm	Key Strength	Limitation	Clinical Use
Rule-based Systems	High transparency and controllability	Limited scalability and adaptability	Coding, compliance, and guideline-driven workflows
Traditional Machine Learning	Efficient with structured features	Requires extensive feature engineering	Classification and moderate-complexity extraction tasks
Transformer-based Models	Deep contextual understanding of clinical language	Resource-intensive; sensitive to domain shift	EHR analytics, phenotyping, and large-scale text mining
Large Language Models (LLMs)	Flexible reasoning, summarization, and interaction	Risk of hallucination; governance and validation needs	Assistive documentation, decision support, research augmentation

7. Emerging Applications of LLMs and Generative AI in Healthcare NLP

Generative AI and LLMs are transforming healthcare NLP by offering systems an opportunity to leave restricted extraction or categorization duties in favor of adaptable, context-appropriate language perception and production (Yu et al., 2025). This is already reflected in their influence on clinical documentation, summarization, question answering, and patient communication, as well as research processes. Simultaneously, their application in health care is limited by safety, reliability, bias, privacy, evaluation, and governance issues, which are even more rigorous than in most other areas (Rabbani et al., 2025). Recent developments in large language models are major extensions of these methodological foundations and they introduce new capabilities alongside new clinical and ethical aspects. Recent research has shown that clinically adapted foundation models with the capacity to absorb heterogeneous medical information and facilitate decision-making processes have emerged, which has led to a paradigm shift that is focused not on task-specific NLP systems but on generalizable, context-aware clinical AI platforms (Joy, 2025). Recent KnowLLM/ACL literature talks about reliable deployment and analysis of LLMs to domain-specific settings (Padiu, Iacob, Rebedea and Dascalu, 2024).

7.1. Clinical Documentation Support and Summarization

LLMs are becoming more popular in writing or revising clinical notes, creating discharge summaries, and summarizing longitudinal patient history (Bednarczyk et al., 2025). The LLMs can generate more coherent stories and change style to various clinical audiences (e.g., clinician-facing vs patient-facing summary) compared to previous summarization systems (Rawan and Saleh, 2026). Practically, the most promising application is as help documentation, writing drafts that clinicians will verify, but not writing autonomous notes (Ahmad, Seo and Jeon, 2025, Jung et al., 2025).

7.2. Conversational Agents for Triage and Patient Engagement

Generative models assist chatbots to gather symptoms, direct patients to relevant care pathways, respond to medication or appointment queries, and provide personalized education (Allouch, Azaria and Azoulay, 2021). Their major strength is that they can maintain a natural, multi-turn conversation and effectively manage patient variability. These systems are, however, highly constrained in healthcare to prevent unsafe advice and to create escalation in the case of significant uncertainty (Rabbani et al., 2025).

7.3. Clinical Question Answering and Information Retrieval Augmentation

LLMs can enhance retrieval-based processes and translation of clinician queries into enhanced search constructions, summarization of retrieved evidence, and the creation of structured responses based on clinical guidelines or internal protocols (Low et al., 2025). This is especially applicable to the situation of decision support, where clinicians require speed of synthesis and not of raw document retrieval (Wang et al., 2025).

7.4. Information Extraction with Less Task-specific Training

LLMs demonstrate potential in the so-called prompt-based extraction, which is the ability to find problems or medications, adverse events, or social determinants, without having a specific model to solve the particular task (Peng et al., 2025). This may shorten the development time, particularly in cases of rare tasks or institutions that do not have an annotated dataset. Practically, the performance is extremely sensitive to prompting, the length of context, and the quality of input

Table 4
Performance comparison of 15 clinical NLP models.

Model	Dataset	Task	Metric	Performance	Reference
BioBERT	BC5CDR	Entity Recognition (Disease/Chemical)	F1-Score	89.3–92.4%	(Sim et al., 2023)
ClinicalBERT	i2b2 2010	Named Entity Recognition	F1-Score	87.2–89.0%	(Kühnel and Fluck, 2022)
BioClinicalBERT	i2b2 2010/2012	NER + de-identification	F1-Score	88–91%	(Wu et al., 2015)
PubMedBERT	BLURB	Biomedical NLP (General)	BLURB Score	81.35–82.91%	(Ao et al., 2026)
ClinicalBioBERT	n2c2 2018	Smoker Status Classification	F1-Score	91–93%	(Naseem, Dunn, Khushi and Kim, 2022)
CheXbert	CheXpert, Radiology	Report Labeling	Accuracy	89–92%	(Abdullah and Kim, 2025)
RoBERTa-MIMIC	MIMIC-III	Phenotyping + Named Entity Recognition (NER)	F1-Score	83–87%	(Tian et al., 2021)
DistilBERT-Clinical	i2b2 2012	Protected Health Information (PHI) Detection + NER	F1-Score	83–86%	(Fraile Navarro et al., 2023)
Med-PaLM	MultiMedQA	Medical Question Answering	Accuracy	82-85%	(Kreimeyer et al., 2017)
GPT-4 Clinical	MedQA / USMLE	Medical Reasoning	Accuracy	86–90%	(Jo et al., 2024)

text (Peng et al., 2024).

7.5. Coding, Billing, and Administrative Automation

Generative models are under investigation to propose medical codes, summarization of documentation to support billing, and prior-authorization writing (Nasser, 2025). These processes are desirable since they are text-intensive and time-consuming; however, they entail solid checking since the presence of minor errors may lead to regulatory and monetary impact (Sangamuang et al., 2025).

7.6. Multimodal and Multimarket Clinical Workflows

More recent progress in generative AI technology has made it possible to use large language models in multimodal clinical settings, where textual data are used with imaging, laboratory values, waveforms, and other structured data (Gupta et al., 2025). LLM is a contextual interface that works in these environments by grounding heterogeneous clinical signals in these settings, instead of defining specialized diagnostic models (Nam et al., 2025). In radiology, multimodal generative systems may aid in report writing by combining the results of computer vision models on the image with clinical history that is obtained through electronic health records. These systems can enhance report coherence and minimize omissions by correlating imaging findings with other findings and the laboratory results with different conclusions and relevant symptoms. Notably, they serve as report drafting or prioritization schemes, the ultimate interpretation of which falls under the mandate of the radiologist (Jandoubi and Akhloufi, 2025). Generative models have potential uses in pathology and oncology to aid in interpretation by synthesizing information in pathology reports, genomic annotations, and past clinical notes. This facilitates the computation of combined summaries that bring out clinically significant biomarkers, treatment implications, and longitudinal disease patterns (Huhulea et al., 2025). Nonetheless, the inconsistency of reporting criteria and the stakes involved in oncologic decision-making require a high level of validation and expert guidance. To provide patient-level clinical summarization, MLLMs have the potential to summarize progress notes, laboratory trends, medication histories, and imaging summaries into brief longitudinal summaries (Nazha et al., 2025). These can be used to enhance handovers, multidisciplinary team meetings, or complicated case reviews. The biggest risk of such applications is the selective omission or misinterpretation of important results, which may lead to the introduction of cognitive bias when clinicians depend excessively on generated summaries. Multimodal workflows create great challenges of governance from a deployment perspective (Khalaf, 2023). The part of the system that makes it difficult to attribute errors can cause errors in one modality to be transmitted in the integrated output. Constrained generation, explicit provenance of sources of information, confidence signaling, and explicit human-in-the-loop validation are therefore usually necessary to enable safe deployment. Multimodal generative AI in healthcare is thus only most advantageous to decision support and efficiency improvements as opposed to independent clinical decision-making (Joy, 2025).

7.7. Validation Protocols for Clinical Deployment

Generative AI and LLMs require explicit minimum validation requirements before clinical deployment. LLM-generated outputs should not be used in clinical documentation or decision support without rigorous validation, since high benchmark performance does not guarantee safe clinical operation and generative models may produce hallucinated, incomplete, or clinically inappropriate content. Deployment frameworks must incorporate technical validation, clinical evaluation, and continuous monitoring to ensure reliability and patient safety (Asgari et al., 2025).

Three minimum requirements are essential. First, human-in-the-loop verification is mandatory for high-risk applications: LLM-generated summaries, diagnostic suggestions, medication recommendations, and discharge documentation must be reviewed and approved by qualified clinicians before use, with documented training, performance monitoring, and records of review processes to mitigate automation bias and prevent clinically significant errors (Moëll and Sand Aronsson, 2025). Second, hallucination detection mechanisms must be implemented, including retrieval-augmented generation (RAG), source-grounded generation, confidence calibration, uncertainty estimation, and automated fact-checking against electronic health records and trusted medical knowledge bases; continuous-learning frameworks integrating structured clinical databases with information-theoretic classifiers can detect factual and reasoning-based hallucinations, reporting hallucination probabilities and achieving AUCs of 0.95–0.96 on medical benchmarks (Zeba et al., 2026). Third, comprehensive audit trails must record prompts, generated outputs, model versions, timestamps, supporting evidence sources, confidence scores, drift detection, inference-level logging, retraining protocols, and clinician modifications to ensure accountability, clinical governance, regulatory compliance (HIPAA, FDA, etc.), transparency, error investigation, and continuous quality improvement (Alu and Oluwadare, 2026).

Prior to clinical use, LLM systems should undergo multiple institutional external validation, prospective testing in clinical workflows, thorough external analysis of hallucination and error rates, clinician-assisted verification processes, and continuous monitoring after deployment. These safeguards guarantee that generative AI can offer safe, reliable, and clinically relevant assistance, rather than decision-making (Artsi et al., 2025).

7.8. Research and Pharmacovigilance

LLMs can greatly accelerate research and pharmacovigilance; however, they cannot substitute epidemiological or statistical analysis. They can be used in literature screening, evidence extraction, cohort definition, and exploratory safety signal detection, but their results should be independently obtained because they run a risk of bias,

misinterpretation, and false associations (Li et al., 2024). LLMs are good at unstructured text in the field of pharmacovigilance, but do not create causality; they need to be combined with structured data and be reviewed by experts (Crisafulli et al., 2025, Kumar, Kaur, Shruti1 and Kaur, 2025). Even though the use of research involves reduced clinical risk in the short term, a transparent governance and a clear distinction between the use of AI in exploratory and confirmatory analysis is required to enforce scientific rigor (Li et al., 2024).

8. Real-World Deployment of NLP in Healthcare: Lessons from Practice

Despite promising technical capabilities, the true value of these systems is determined by their performance in real-world healthcare environments, where implementation realities shape outcomes. Although many case studies testify to the possibility of NLP in healthcare, they hold no separate technical merit but serve as a predictor of the lessons learned regarding implementation, clinical outcome, and restriction. Instead of providing a collection of unrelated application overviews, this section combines practical applications of NLP based on common clinical goals, and the cross-cutting insights are used in future translations (Garg and Gupta, 2023).

8.1. Classification of Medical Specialties

NLP has been applied in various health care environments to derive ordered information about unstructured clinical stories as well as to summarize longitudinal patient histories and demonstrate a steady advantage of increasing information access and decreasing clinician workload, especially in large volume settings like emergency and intensive care units (Carriere et al., 2021). Nevertheless, in real life, performance usually lags behind benchmark findings in controlled studies. A lack of consistency in documentation practices, missing documentation, and time ambiguity handicap robustness are problems. Effective roll-outs are thus favored to focus on limited operations, people-in-the-field validation, and incremental inference into working processes, instead of a totally automated choice (Van Bulck, Reading Turchioe, Topaz and Song, 2025).

8.2. NLP in Diagnostic Support and Clinical Interpretation

NLP has been used in the support of diagnostic reasoning in areas like imaging reporting, cancer narratives, and specialty consultation. In reality, these systems are best applied as a support tool in raising the pertinent findings or patterns and not as a self-directed diagnostic machine (Casey et al., 2021). There is data on deployed systems that shows interpretability and alignment with clinical reasoning have a strong effect on adoption. Models with traceable outputs are more easily accepted than black box predictions, even when the latter are more accurate in isolation (Wang and Luo, 2025).

8.3. Mental Health and Behavioral Monitoring

The discussion of language patterns as a method of mental health diagnosis and monitoring is one of the most promising practical applications of NLP (Malgaroli, Hull, Zech and Althoff, 2023). Clinical notes, patient communications, and online interactions have been identified as having linguistic predictors of depression, anxiety, and suicide risk using NLP systems. Although these applications are promising, there are increased ethical and methodological issues with them (Montejo-Ráez et al., 2024). Indicators based on language are very sensitive to context, culture, and personally express, making it more likely to be biased. Consequently, systems deployed focus on screening and risk flagging, and clinician oversight is still a necessity (Ghorbian and Ghobaei-Arani, 2025).

8.4. Adverse Event Detection and Patient Safety

NLP has been commonly investigated to identify adverse drug events and safety signals in clinical text. In comparison to the conventional reporting systems, NLP facilitates enhanced surveillance due to the wider capture of events that are recorded orally in clinical stories (Murphy et al., 2023). But practice has shown that there is a trade-off between sensitivity and specificity, and that it is difficult to tell the difference between a true cause-and-effect relationship and a mere coincidence. Organizational implementations are thus effective when they combine the NLP output with the knowledge of pharmacovigilance and formal data sources to reduce false alerts and clinical trust (Young, Luz and Lone, 2019).

9. Case Studies

9.1. NLP in Artificial Intelligence

AI is associated with radical changes, including medical imaging, and some areas that affect or impact the disease detection process and the surgical planning process. In one article it demonstrates development both on the technological and clinical level, as summarized through Figure 8(1-6). Figure 8(1) shows that AI-in-medical-imaging research expanded quickly in 2023; not only because of more interest fueling many innovations, but also because of the breadth the architecture approaches such as CNNs, transformers, GANs, and hybrids, which accommodate challenges in image capture, analysis, and interpretation. Figure 8 is devoted to the architecture of the transformers, illustrating how vision transformers divide images into patches to perform positional encoding and use attention mechanisms. Figures 8(3) and 8(4) introduce such developments as hierarchical partition in SWIN Transformer and shifted-window attention that enhance scalability to medical segmentation tasks. Figure 8(5) and 8(6) provide an applied angle, where Figure 8 illustrates GAN-based frameworks in developing realistic synthetic medical images to address the issue of data scarcity and privacy restrictions, and in Figure 8(6), the focus areas of research are mapped, with strong dominance in the realm of lung issues brought by COVID-19 and lung cancer cases, followed by other vital organs (Pinto-Coelho, 2023).

9.2. NLP for Mental Health Prediction

NLP is at the forefront of predictive analytics for mental health, particularly when it comes to digital psychiatry and remote care. To evaluate psychological risk and emotional well-being, contemporary systems use textual data from phone transcripts, emails, online therapy discussions, patient diaries, and social media. To identify language indicators like negative attitude, passive speech, decreased complexity, and frequent use of personal pronouns, often associated with depression, PTSD, or suicidal thoughts, tools like CogStack and CLPsych (Computational Linguistics and Clinical Psychology) are utilized as shown in Figure 9. For instance, by examining language from therapy notes and patient interactions, research conducted at Veterans Affairs hospitals employed NLP to identify veterans who were at high risk of suicide and to initiate real-time treatments (Zhang, Schoene, Ji and Ananiadou, 2022, Ibrahim, Li, Patel and Katapally, 2025, Malgaroli, Hull, Zech and Althoff, 2023).

Conversational NLP is also being used by smartphone apps (such as Wysa and Woebot) to provide cognitive behavioral therapy (CBT) techniques and gauge user mood. Adding a multimodal element to mental health tracking, a recent study also supports the use of NLP with voice-based emotion recognition for individuals with anxiety disorders or dementia. In addition to predicting relapse, these tools enable doctors to track progress over time and tailor therapy routes (Barath, Murali and Pavithra, 2025, Pothuri, 2024). The Latest Case Studies are discussed in Table 2. Over the past decade, there has been a significant increase in

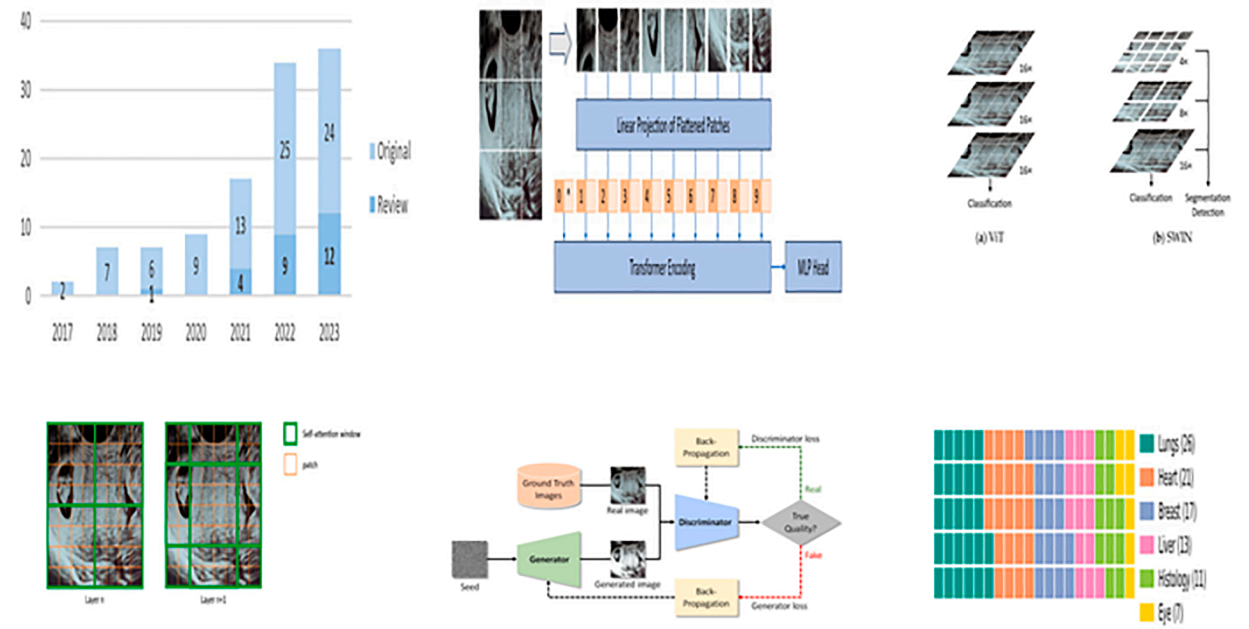


Figure 8. The figure illustrates the distribution of knowledge utilization techniques in biomedical text mining. Ontologies emerge as the dominant method, accounting for 55.9% of usage, followed by customized dictionaries and pretrained embeddings, each at 14.7%, while other methods and ML-BERT contribute 8.8% and 5.9%, respectively. Among the ontologies, UMLS is the most frequently used resource, appearing in the highest number of studies, followed by SNOMED-CT, MeSH, ChEBI, and UniProt. Overall, the figure highlights the central role of ontologies, particularly UMLS, in supporting knowledge-driven biomedical text mining approaches compared to other methods (Pinto-Coelho, 2023).

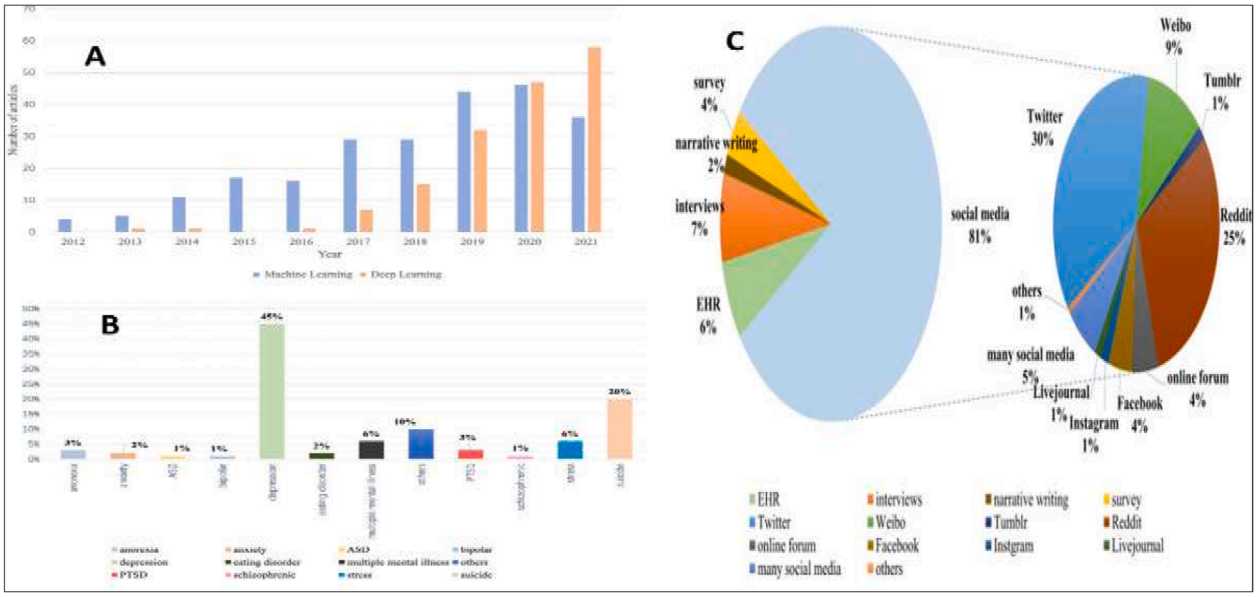


Figure 9. (A) Yearly number of published articles on machine learning and deep learning for mental health from 2012 to 2021, showing rapid growth in deep learning after 2018. (B) Distribution of mental health topics studied, with depression and suicide as the most frequently addressed conditions in published articles. (C) Sources of data in mental health research highlight the dominance of social media platforms, especially Twitter and Reddit, over traditional sources (Zhang, Schoene, Ji and Ananiadou, 2022).

published research articles applying machine learning and deep learning techniques to mental health, with deep learning studies accelerating particularly after 2018 (Figure 9(A)). Much of this research has centered on conditions like depression and suicide, which together make up the majority of studied topics (Figure 9B). Additionally, the bulk of these studies draw upon data sourced from social media platforms such as Twitter and Reddit, reflecting the growing importance of online interactions in mental health research, while traditional sources like interviews and surveys make up a smaller portion (Figure 9C) (Zhang, Schoene, Ji and Ananiadou, 2022).

9.3. *Advancing ADE Detection with NLP—Trends, Networks, and Annotation Strategies*

From 2011 to 2021, research on NLP-based Adverse Drug Event (ADE) detection in electronic health records exhibited rapid growth,

especially in 2019–2020, with most publications originating from the United States (Figure 10(A)). These efforts drew on interdisciplinary expertise from medical informatics, clinical medicine, and computer science, yet demonstrated relatively limited international collaboration. Network analysis (Figure 10(B)) revealed that ADE detection research spans a wide range of clinical specialties, including oncology, critical care, emergency, and psychiatry, and involves collaborations among key researchers who often contribute across several domains, fostering multidisciplinary advances improving the applicability of NLP models to diverse patient populations. At the technical core of this research, strategies for annotating ADEs in clinical text diverged: some studies labeled ADE as a standalone entity. In contrast, others modeled ADE as a relationship between a drug and a symptom or sign (Figure 10(C)). Given the complexity of clinical language and the need for accurate causal interpretation, treating ADEs as relationships (rather than isolated entities) proved more effective, aligning machine learning outputs more closely with clinical reasoning and enhancing the precision of automated ADE detection (Murphy et al., 2023).

Table 5 summarizes and can be used to demonstrate the application of NLP frequently in accordance with ML in healthcare data of various types to enhance the analysis, classification, and clinical decision-making. One of the key topics is the use of NLP in the unstructured text coding, where NLP-based approaches in combination with feature selection or deep learning, are used to predict medical specialties, breast lesions in cytopathology reports, and multi-label clinical texts (Sim et al., 2023). These methods illustrate the ability of NLP to encode free-text reports into meaningful categories that are organized and structured, enhancing the organization, interpretability, and subsequent utilization of the data in the electronic medical record (EMRs) (Velupillai et al., 2018). Some of the studies go beyond what can be analyzed by text to the case of integrating text with image processing in the medical field. Through their convergence method of applying NLP and image analysis techniques, such works demonstrate that multimodal data can be used to complement each other to better interpret multi-faceted clinical data (Huang et al., 2024). In addition, the application of the advanced deep learning architecture, including capsule networks, is also demonstrated as a means of improvement of medical text classification, being more successful than its classical counterparts (Khalate, Gite, Pradhan and Lee, 2024). The table also contains the studies dedicated to the information extraction of unstructured healthcare records, which prove the efficiency of the NLP pipes in retrieving clinically relevant information in free-text and scanned EMR documents. These strategies go a long way to enhance the accessibility and retrieval of data, solving long-term problems that are related to legacy and non-digitized records. The other significant category of works focuses on the analysis of medical images, in which the ML methods are used to extract features, conduct segmentation, registration, and detection. These articles demonstrate that more sophisticated ML designs can be used to improve the analysis and interpretation of medical images to facilitate a more accurate diagnosis and clinical evaluation. Other studies that complement original research show the value of NLP and ML in enhancing EMRs by certifying domain-specific language models, e.g., those that predict conditions based on radiology notes, and making them scalable and multi-site. Lastly, the table consists of detailed reviews of NLP being applied to specialized fields/areas, including clinical neuroscience, psychiatry, and radiology. All these reviews show the increasing effect of NLP in deriving insights in EMRs, research support, and improved patient care. In general, the table points to the multifaceted application of NLP in textual, imaging, and multimodal healthcare data, which is important in the development of automated analysis, clinical documentation, and decision support systems.

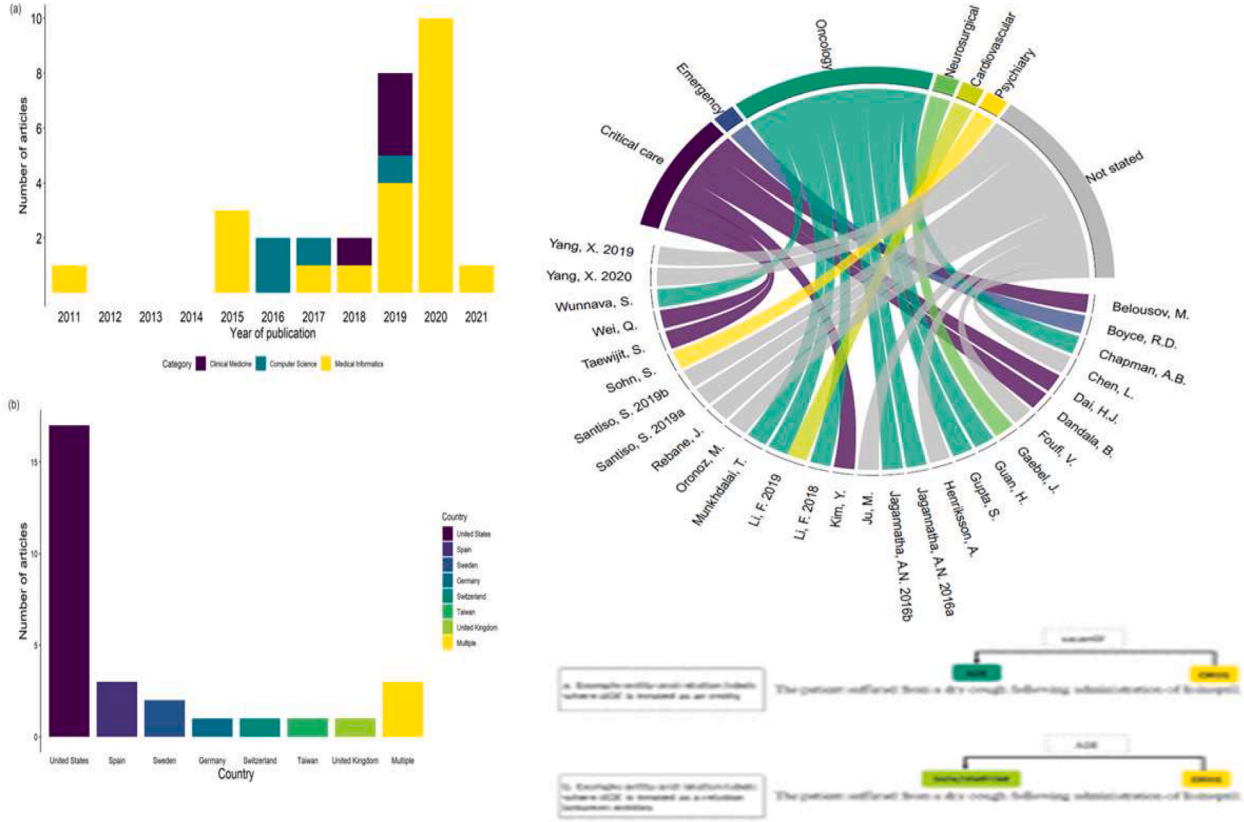


Figure 10. Trends in ADE-NLP publications (2011–2021) by category and country, showing a research surge in the US, especially in medical informatics. Chord diagram connecting authors and clinical subfields in ADE-NLP, highlighting multidisciplinary collaborations, especially in oncology and critical care. Comparison of annotation strategies for ADEs in NLP: treating ADE as an entity versus as a relationship between drug and symptom entities (Murphy et al., 2023).

Table 5
Important case studies related to the use of NLP in health science.

Sr No.	Use of NLP	Objectives	Methodology	Key findings	Reference
1	Unstructured text coding	To classify medical specialties in textual reports	A combination of NLP and feature selection techniques	Classifies medical specialties in textual reports, enhancing data organization in EMRs	(Eguia et al., 2024, Roy et al., 2021)
2		To classify breast lesions in cytopathology reports	Multilevel and multiclass NLP approaches	Implements multilevel and multiclass classification strategies for breast lesions in free-text cytopathology reports	(Abdelwahap et al., Edin et al., 2023)
3		To justify multi-label text classifications in healthcare	Analyzing the rationale behind multi-label classifications using NLP	Addresses the rationale behind multi-label text classifications in healthcare, enhancing understanding of NLP applications	(Abdelwahap et al., Xian et al., 2024)
4		To explore text and image processing in medical applications	Investigating the combined use of text and image processing techniques	Enhances understanding of how these technologies can be integrated in medical settings	(Hossain et al., 2023, Ji, Chen and Jiang, 2020)
5		To improve medical text classification	Utilization of an improved capsule network	Demonstrates enhanced performance in medical text classification	(Li et al., 2023, Ye, Woods, Jordan and Starren, 2024)
6		To extract information from unstructured healthcare records	Application of NLP for data extraction	Shows a different approach to handling unstructured data in healthcare records	(Amirahmadi, Ohlsson and Etminani, 2023, Luo et al., 2020)
7	Medical image analysis	To survey ML applications in medical image analysis	Conducted a comprehensive survey of existing techniques	Provided an extensive overview of ML in medical image analysis	(Xian et al., 2024, Jerfy, Selden and Balkrishnan, 2024)
8		To extract multi-scale features in medical imaging	Developed PLU-Net for feature fusion	Showcased improved feature extraction capabilities	(Li et al., 2023, Orynbay et al., 2025)
9		To analyze medical images with ML techniques	Utilized various ML techniques for image analysis	Demonstrated improved analysis and interpretation of medical images	(Ahmed et al., 2024, Luo et al., 2020)
10		To improve feature extraction for image analysis and detection	Applied ML techniques for feature extraction	Enhanced detection and analysis of images	(Ball and McBeth, 2021, Johri et al., 2020)
11		To analyze medical images using image registration and segmentation	Focused on feature-based segmentation techniques	Provided insights into effective image analysis strategies	(Pothuri, 2024, Haas et al., 2021)
12	NLP and ML for improvements in EMR	Extract information from scanned EHR documents using deep learning-based NLP	Development of a deep learning NLP data pipeline	Improved data retrieval methods from scanned EHR documents	[14] [20]
13		Investigate the integration of text and image processing in medical contexts	Examination of combined medical text and image processing techniques	Enhanced EMR analysis through integrated text and image processing	(Hossain et al., 2023, Damera et al., 2025)
14		Validate the effectiveness of RespBERT in identifying ARDS from radiology notes	Multi-site validation of a NLP algorithm	Successful identification of ARDS in EMRs using RespBERT	(Klug et al., 2024, Ferreira et al., 2023)
15		Review NLP applications in clinical neuroscience and psychiatry	Comprehensive review of NLP use cases in relevant fields	Potential of NLP in enhancing EMR analysis and patient care in neuroscience and psychiatry	(Kumah, Osei-Kesse and Anaba, 2017, López-Úbeda, Martín-Noguerol, Aneiros-Fernández and Luna, 2022)
16		Discuss the role of NLP in radiology research and clinical applications	Exploration of NLP technologies in radiology	Significant impact of NLP on radiology research and clinical applications in EMR analysis	(Thummak, Uppor and Wannarit, 2023, Siebra, Kurpicz-Briki and Wac, 2024)

10. Challenges and Constraints Shaping NLP Deployment in Healthcare

Despite the rapid maturation of NLP techniques, there is a mixture of data, methodological, clinical, ethical and infrastructural limitations that limit the translation of NLP techniques into routine clinical practice (Okwor et al., 2024). More importantly, these issues are not just on paper but directly have an impact on model accuracy, robustness, generalizability, interpretability and trust by clinicians (Maleki Varnosfaderani and Forouzanfar, 2024). Insights from practical deployment highlight a range of cross-cutting challenges that must be addressed to ensure safe, reliable, and scalable use of NLP in healthcare.

10.1. Data Heterogeneity, Domain Shift, and Performance Degradation

Clinical documentation practices are known to be quite diverse among institutions, specialties, and even among individual clinicians (Herrington and Wang, 2023). Variations in terminology, abbreviations, and structure of notes and clinical culture create a domain shift that significantly causes negative model performance outside of its training space. Empirical experiments always show that models that are highly accurate in benchmark data tend to exhibit significant recall and precision degradation when used in novel settings (Yang et al., 2025). This makes scaling less scalable and more expensive to maintain as models might need re-fine tuning, adaptation to particular sites or constant recalibration. Generally speaking, generalizability can be more significant than marginal gains when benchmark performance is involved (Serbanati, 2020).

10.2. Linguistic Ambiguity, Temporality, and Clinical Context

Clinical text is ambiguous and situational in nature. Negation (no evidence of disease), uncertainty (possibility of being infected), speculation (rule out stroke), and temporal changes (history of cancer), need to be interpreted appropriately to prevent clinically meaningful adverse errors (Gravholt et al., 2024). Contextual understanding failures are overrepresented in high-risk tasks, including adverse event detection or diagnostic support, where the false signals or false alerts can be caused by misinterpretation. The even state of the art models are susceptible to such obstacles, especially in dealing with long clinical histories or

discontinuous records across multiple encounters (Yadav, Patel and Shah, 2021).

10.3. Label Quality, Annotation Bias, and Ground Truth Uncertainty

Quality annotated data are necessary in supervised NLP, but clinical annotation is both expensive and subjective, and it may also be impacted by local practice patterns (Sylolypavan, Sleeman, Wu and Sim, 2023). Documentation habits are frequently reported in labels, which do not represent clinical reality, instead adding noise and bias to training data. These restrictions influence the performance of the model and assessment (Zhang et al., 2023). Lack of consistency in the standards of annotation makes cross-study comparison more difficult and label uncertainty can lead to an illusion of performance in development and less reliable in practice. Due to this, the performance metrics need to be taken with a lot of caution and they should be supplemented with the qualitative clinical validation (Siddique et al., 2024).

10.4. Data Sparsity, Class Imbalance, and Rare Events

Clinical concepts of interest, including infrequent diseases, drug adverse effects, or safety warnings, are poorly represented in clinical text corpora. Extreme imbalance in classes makes the model insensitive and prone to false-negative, especially in safety-critical models (Santiso, Casillas and Pérez, 2019). Although these problems can be reduced to some extent with the help of data augmentation and transfer learning, rare-event detection has become one of the limitations that cannot be completely overcome. This can frequently require strict thresholds in deployment; or hybrid systems based on a combination of NLP outputs and rule-based filters and human inspection (Cartus, Samuels, Cerdá and Marshall, 2023).

10.5. Privacy Regulations and Constraints on Model Development

Data protection laws are tough and they place needed yet significant limitations on health care NLP. The restriction of data sharing minimizes the diversity of the data sets, limits the multi-site validation and limits reproducibility (Conduah, Ofoe and Siaw-Marfo, 2025). Most of the well-performing models are based on proprietary datasets, which are not retestable. These limitations increase the domain shift and bias, with models being trained on small populations and environments. As far as deployment is concerned, privacy-sensitive solutions (e.g., federated learning) are promising but add an extra technical and organizational burden (Chen et al., 2025).

10.6. Bias, Fairness, and Equity Implications

Clinical text mirrors existing healthcare access disparities, disparities in healthcare diagnosis procedures, and in clinician patient relationships. NLP models that are trained using this type of data are associated with the risk of reinforcing or enhancing inequities based on demographic groups (Cross, Choma and Onofrey, 2024). Such bias can be in the form of dissimilar error rates, less sensitivity to underrepresented groups of people, or systematic misinterpretation of language styles. Notably, effects can be mostly unseen in terms of aggregate performance measures and only become apparent in practice (Chinta et al., 2025). To tackle subgroup analysis, clear reporting, and continuous monitoring as opposed to periodic evaluation is essential.

10.7. Interpretability, Explainability, and Clinical Trust

Interpretability is a requirement of adoption in healthcare. Clinicians do give more credibility to systems that offer traceable outputs, e.g., highlighted text, extracted entities, or summarized evidence, than do opaque predictions (Ennab and McHeick, 2024). Very complicated models can be more effective but will also be opposed in case their reasoning is inaccessible to questioning or description. Consequently, there are frequent trade-offs between accuracy and transparency in deployment decision-making, especially in decision-support situations (Muhammad, Salman and Bendechache, 2025).

10.7.1. Explainable AI Techniques for Clinical NLP

Explainable AI (XAI) methods improve the transparency of clinical NLP models by explaining how predictions are made. Common techniques include SHAP, LIME, attention visualization, saliency mapping, and transformer-based attribution methods, which help identify the features or tokens that most influence model decisions. The explainability methods described include SHAP, LIME, attention visualization, saliency mapping, and transformer-based attribution techniques for interpreting model predictions.

10.7.1.1 SHapley Additive exPlanations (SHAP). SHAP is a game-theoretic framework for understanding the predictions of any ML model and measures feature importance and prediction contributions. SHAP is inspired by the Shapley values used in cooperative game theory, which are related to optimal credit allocation, and offers a unified framework for understanding the model's predictions (Ponce-Bobadilla et al., 2024). SHAP is the only explainability method that meets the local accuracy, missingness, and consistency properties, which guarantees mathematically sound explanations. SHAP values offer local explanations for each prediction, and it also enables users to understand the importance of each feature globally by aggregating SHAP values across datasets. Additionally, SHAP is directional, showing whether each feature increases (+) or decreases (-) predictions relative to the baseline (Hur et al., 2025).

SHAP analysis offers actionable advice for interpreting predictions from supervised ML models in drug development, uncovering important biomarkers and clinical characteristics that influence predictions (Muhammad, Salman and Bendechache, 2025). It is applied in clinical decision support systems in practical SHAP tutorials, and the force plots explain individual patient risk prediction (Cakiroglu, Korkmaz, Bekdaş and Nehdi, 2026). SHAP-based insights serve as a cornerstone for deciphering relative significance of variables in complex ML models across domains including healthcare. In temporal feature importance analysis, SHAP values reveal which clinical measurements contribute most to disease prediction over time. SHAP values explain individual samples using force plots, providing measures of geochemical/clinical attribute importance in diagnostic tasks (Antonini et al., 2024). For binary classification, SHAP importance charts display average absolute SHAP values for each feature, sorted from the highest to lowest influence.

SHAP offers four calculation methods tailored to different model types. Tree SHAP is the fast, exact method for tree-based models including Random Forest, XGBoost, LightGBM, and CatBoost [Qlik]. Linear SHAP computes SHAP values for linear models such as Logistic Regression, Linear Regression, Lasso, and Elastic Net [Qlik]. Kernel SHAP is model-agnostic and works with any ML algorithm using an approximation method with weighted sampling, making it universally applicable but computationally more intensive. Kernel SHAP can be expensive for large datasets. Deep SHAP is specifically designed for neural networks and deep learning models, serving as an extension of DeepLIFT optimized for deep architectures. SHAP values may be unstable when features are highly correlated. and requires training for clinicians (to understand SHAP plots) (Ponce-Bobadilla et al., 2024).

10.7.1.2 LIME (Local Interpretable Model-agnostic Explanations). LIME is an XAI technique that explains individual predictions by approximating complex black-box models with interpretable surrogate models locally around each prediction. LIME focuses on local explanations at the individual patient-level, making it particularly valuable for clinical NLP where understanding why a specific patient was classified as high-risk is

critical (Houdou et al., 2023).

It is model-agnostic, working with neural networks, random forests, SVMs, and transformers. LIME produces interpretable explanations using sparse linear models or decision trees and has built-in text support with word-level explanations showing which words drove predictions (Hassan, Abdulkadir, Zahid and SM, 2025). LIME highlights words/-phrases driving individual patient risk predictions, enabling clinicians to validate model reasoning. For text classification, LIME perturbs text by removing/masking words and measures prediction probability changes to identify influential features. LIME combined with SHAP provides enhanced interpretability for ML models, with LIME offering faster local explanations and SHAP providing theoretically-grounded global importance (Vimbi, Shaffi and Mahmud, 2024). LIME is preferred for individual prediction explanations in clinical settings where clinicians need to understand specific patient-level decisions. Key limitations include instability (explanations vary across perturbations), local-only scope (may not reflect global behavior), and fidelity tradeoff (balancing simplicity vs. accuracy). Despite limitations, LIME supports text, tabular data, and images with scikit-learn integration.

10.7.1.3 Attention Visualization. Attention visualization highlights influential words/tokens by displaying attention weights in transformer models, showing which parts of the input the model focuses on when making predictions. Attention mechanisms compute the attention score between input tokens by applying a Query (Q), Key (K) and Value (V) vector to focus on relevant parts of the input. These weights are shown as heatmaps or as highlighted text in the visualization making the model's focus interpretable (Ji et al., 2021). Key features are built-in interpretability (no post-hoc computation needed), token-level focus (word-by--word importance), multi-head attention (see various attention patterns) and dynamic focus (brightness indicates attention weight intensity) (Liao, Gao and Zhang, 2025).

In clinical NLP, attention mechanisms are universally used in contemporary DL models, significantly improving performance in clinical text classifications. BERT, GPT-2, BART, and ClinicalBERT support are available for visualization with BertViz. Clinical NLP systems visualize the weights of attention to the clinical terms that are associated with disease predictions, allowing the clinician to validate the system. The advantages are that it doesn't require model training, is easy to visualize for clinicians, can capture long-range dependencies, and can compute rapidly with native model weights (Ramos and Collison, 2025). For implementation, BertViz offers visualization of attention patterns for clinical text using BERT, GPT-2, BART, and ClinicalBERT. But, there are some drawbacks to attention visualization. Attention does not equal importance since high attention weight doesn't always mean a feature is important. Attention patterns can be misleading where attention patterns may not correspond to actual causal relationships. It needs complementary methods (such as SHAP/LIME) to provide strong explanations. It may be difficult to interpret multiple attention heads because of complexity (Kim et al., 2020).

For implementation, BertViz and similar tools support BERT, GPT-2, BART, ClinicalBERT, and other transformer models for visualizing attention patterns in clinical text through simple Python code using the "hea" view function.

10.7.1.3 Saliency Mapping. Saliency mapping is a gradient based XAI approach which is used to determine the most important regions of an input, such as words in text or pixels in images, that contribute most to a model's prediction. It calculates a "saliency map" using gradients of the change in output for a change in each of the inputs, to find the significance of each input region according to whether it is a visual feature or a number (Martin et al., 2026). This principle is based on computing the gradients of the output of the model with respect to the input in a forward pass and determining the sensitivity of the output to the input, to determine which features are most important for the model's prediction. These gradients can be used with methods such as Gradient-weighted Class Activation Mapping (Grad-CAM) to produce heatmaps of key areas.

Saliency maps are widely used for explaining deep learning models in medical applications. They are used as biomarkers in 3D medical imaging classification tasks. They show the impact of individual pixels or words for disease detection and clinical text classification. Gradient-based analysis is based on the changes in gradients at training epochs for feature importance estimation. Benefits include speedy importance scores, ease of use for any differentiable model, such as CNN or transformer, ease of debugging and improving the model, and displaying important clinical terms at the token level in NLP (Aysel, Cai and Prugel-Bennett, 2025). However, limitations include potential noise or highlighting irrelevant features in complex architectures, explaining *where* features matter but not *why*, not reflecting true causal relationships, and requiring combination with other XAI methods (LIME, SHAP) for robust explanations. Implementation works with CNNs, transformers, and other differentiable models, where gradient-based feature attribution methods provide a simple implement ability for deep neural networks (van Zyl, Ye and Naidoo, 2024).

10.7.1.4 Explainable Transformers. To overcome this challenge of interpretability in deep transformer models, explainable transformers utilize specific attribution techniques to track the model's internal processes and explain the factors behind its predictions. Attention rollout, integrated gradients and layer-wise relevance propagation (LRP) are three widely used methods for explaining transformers (Rahnfeld et al., 2025). Attention rollout is a recursive algorithm that remedies attention explainability for deeper layers by aggregating attention weights across all layers to show information flow from input to output, with gradient attention rollout extending this by multiplying attention with target class gradients for effective vision transformer and class-specific explainability (Tian, 2025). Integrated gradients (IG) computes the integral of gradients along the path from baseline to actual input to attribute predictions, with the context-aware layer-wise integrated gradients (CA-LIG) framework computing layer-wise gradients within each transformer block and fusing token-level attributions with attention gradients to create context-sensitive attribution maps providing more faithful, semantically coherent visualizations across sentiment analysis (Murali and Mazumder, 2026), document classification, and image classification tasks; layer-wise relevance propagation (LRP) propagates relevance scores backward through the network by redistributing activation values, outperforming gradient- and perturbation-based methods with computational complexity while providing per-neuron importance scores, reformulated to handle positional encoding (rotary, learned, absolute PE), with LXT extending LRP to handle attention layers in large language models and vision transformers (Tehsin, Alshaya, Bouchelligua and Nasir, 2025).

SHAP, LIME, attention visualizations, saliency maps, and explainable transformer methods collectively enhance the interpretability of clinical NLP models, increasing transparency, building clinician trust, and promoting adoption of AI-driven healthcare systems.

10.8. Workflow Integration and Human–AI Interaction

Clinical impact cannot be based only on technical performance. The NLP systems should not disrupt the workflow and should reduce the cognitive load and provide the information at the appropriate time in clinical terms (Abd-Alrazaq et al., 2025). Poorly integrated systems can cause alert fatigue, impair documentation practices or be bypassed (Cunha Reis, 2025). Effective implementations focus on human-in-the-loop implementations, in which NLP supports clinical judgement and is not intended to substitute them. Follow-up systems make systems adapt to varying clinical practices (Aravazhi et al., 2025).

10.9. Infrastructure, Resource, and Sustainability Constraints

Deep NLP implementation needs both computational infrastructure and data engineering skills and generally long-run maintenance strategies (Li, Fu and Python, 2025). Healthcare environments with limited resources might not be able to accommodate large-scale models or retraining regularly (Gao and Adnan, 2025). Software obsolescence, changing regulatory demands and altering clinical guidelines are sustainability issues. These are some of the factors that should be taken into consideration in initial stages of system design to prevent short lived or non-scalable implementations (Peykani, Ramezanlou, Tanasescu and Ghanidel, 2025).

10.9.1. NLP in Low-Resource and Multilingual Healthcare Settings

Despite significant advances in clinical NLP, progress has been concentrated primarily in English and other high-resource languages, leaving many healthcare systems constrained by limited annotated datasets, inadequate multilingual medical terminologies, and insufficient computational resources. A major limitation is the scarcity of annotated clinical corpora: most public datasets and benchmarks concentrate on English and a few high-resource languages, restricting model development and evaluation for underrepresented populations and perpetuating performance disparities (Shinohara and Kawazoe, 2026). Recent studies underscore the persistent lack of multilingual clinical datasets and the urgency of cost-effective annotation strategies to support low-resource medical NLP. The absence of standardized medical terminology across languages hampers semantic interoperability, information extraction, and cross-lingual knowledge transfer, while code-switching, transliteration, and non-standard orthography further complicate tokenization and parsing. Linguistic barriers between providers and patients negatively affect diagnostic accuracy and communication (Noll et al., 2023). It is constrained by limited computational infrastructure, poor internet connectivity, and restricted access to high-performance computing in many healthcare systems. Stringent privacy regulations, ethical concerns, and fragmented governance frameworks often prevent cross-institutional data sharing, impeding the creation of robust, generalizable models. Collectively, these challenges reduce scalability and contribute to inequitable access to AI-enabled healthcare across diverse linguistic and socioeconomic settings (Shahrour, Junejo and Al-Said Ahmad, 2026).

Several methodological advances have emerged to address NLP challenges in low-resource and multilingual healthcare settings. Pretrained multilingual encoders (e.g., XLM-R, mBERT) and more recent cross-lingual LLMs provide strong zero-shot or few-shot transfer for many tasks. Cross-lingual transfer can be improved by target-language continued pretraining on unlabeled in-domain text and vocabulary adaptation (Conneau et al., 2020). ClinicalBERT and Bio+Clinical models adapted for English can be a pattern for creating local-language clinical encoders via continual pretraining. Multilingual clinical NLP shared tasks (e.g., MEDDOCAN-style deidentification tasks, multilingual NER challenges) and regional consortia accelerate dataset creation and benchmarking. Regional datasets and benchmarks include (Mulyar, Uzuner and McInnes, 2021): MIMIC-III / MIMIC-IV (English clinical records) (as an adaptation model, not a direct multilingual resource), I2B2/VA challenge datasets (English), n2c2 and MedNLP shared-task outputs, region-specific effort includes Spanish clinical corpora (e.g., IULA-Speech/Clinical Spanish resources), Chinese clinical notes corpora, and initiatives for Indic languages (recent,workshops and datasets for Hindi, Tamil, Bengali) including multilingual biomedical corpora like mC4 and CC-100 for continued pretraining (Johnson et al., 2023).

Build or adopt cross-lingual benchmarks for core clinical tasks and validate models rely on local datasets; run fairness audits by language, dialect, demographic group, and setting to detect harmful failure modes; include explainability, uncertainty estimates, and clinician review for high-risk tasks; favor compute-efficient methods (distillation, pruning, quantization, adapters) for on-premise or edge deployment; and enforce strict de-identification, data governance, and legal review for multilingual corpus creation (Ahmed et al., 2026). Collaborative consortia of hospitals, universities, and government agencies create de-identified, multilingual corpora with standardized annotation schemas and metadata; develop standardized cross-lingual clinical benchmarks that capture regional variation, code-switching, and real clinical tasks and run shared tasks to improve reproducibility; invest in annotation-efficient pipelines that combine active learning, weak supervision, and semi-supervised pretraining to minimize expert labeling; prioritize lightweight, adaptable models through adapters/LoRA, distillation, and compact deployment toolkits suitable for local or edge use; conduct clinician-in-the-loop and sociotechnical studies to assess workflow fit, adoption barriers, and safety across cultural contexts; and advocate targeted policy and funding to support multilingual dataset creation, local training/hosting infrastructure in LMICs, and ethical frameworks that protect privacy while enabling research (Oliveira et al., 2022).

10.9.2. Standardized Evaluation Frameworks for Clinical NLP Systems

In addition to addressing resource and deployment challenges, successful clinical NLP systems must also be translated into healthcare practice, and standardized frameworks for evaluating clinical NLP systems need to be developed that go beyond technical evaluations of algorithms to include workflow integration, interpretability, safety, and clinical utility. A number of algorithmic metrics like accuracy, precision, recall, F1-score, and AUROC are used for the current evaluation of clinical NLP systems. These tools measure predictive accuracy but are not the same as being clinically useful, assessing patient safety, are interpretable, or are integrated into the workflow. Improved evaluation frameworks should have a multi-dimensional perspective that encompasses technical performance, clinical utility, safety assessment, explainability, fairness, and operational feasibility. Standardized evaluation measures would enable more accurate comparison of NLP paradigms, and would help to bring research models into the real world of healthcare more quickly (Jiang et al., 2026).

Accuracy, precision, recall, F1-score, and AUROC are well-known metrics for clinical classification problems; in clinical text generation and summarization, BLEU and ROUGE scores are used. The metrics offer a common way to evaluate model performance and enable comparisons between various NLP paradigms and datasets. For instance, an NLP system was developed to identify key radiological findings (intracranial haemorrhage) and raise alerts to clinicians in a timely manner. It was trained and initially tested on internal data with good F1 score 0.92, precision 0.94, recall 0.90 and AUROC score 0.96, after which it was tested at another hospital, and the operating threshold was set to focus on sensitivity (Lyu et al., 2026). External validation revealed 12% false negatives in a high-acuity cohort; threshold adjustment reduced false negatives to 3% but increased false positives and alert burden requiring clinician triage. Post-deployment, time-to-notification improved but clinician alert fatigue increased. High F1/AUROC masked a safety issue prioritizing recall for high-risk. In another clinical study, a multilingual NER model was developed to extract medications and diagnoses from clinical notes in English, Spanish, and a low-resource regional language.

The model was trained on multilingual clinical data and initially evaluated using macro-F1 (0.84), micro-F1 (0.88), and AUROC (0.91); stratified evaluation exposed poor performance on the low-resource language (F1 = 0.52), especially for code-switched and transliterated text. The researchers then applied continued pretraining on unlabeled local clinical text, added adapter modules for the low-resource language, and collected more expert annotations via active learning. Performance for the low-resource language improved to F1 = 0.71 but remained lower than for English; the researchers implemented a clinician review for low-confidence extractions in that language until performance improved. Macro-averaged metrics hid dangerous subgroup disparities fairness audits and stratified evaluation are essential for equitable deployment (Fang et al., 2026).

Clinical NLP evaluation must go beyond algorithmic metrics to assess real-world impact and must measure clinical utility (diagnostic concordance, time saved per patient, reduced clinician burden, recommendation acceptance, and patient outcomes like readmission/adverse-event reduction) (Tamang et al., 2023); assess safety (hallucination rate, incorrect content, clinical error rate, robustness under noisy records, and bias across languages/demographics) (Lyu et al., 2026); evaluate interpretability (attention visualization, source attribution, confidence calibration/ECE, and clinician trust via user surveys); and quantify workflow integration (EHR compatibility, response time/latency, adoption/usage frequency, and additional clicks/time burden). Together, these metrics align technical performance with clinical needs, safety, fairness, and deployment feasibility.

10.9.3. Uncertainty-Aware and Adaptive Clinical NLP Systems

As clinical NLP systems are increasingly integrated into decision-support workflows, the ability to manage uncertainty and maintain robust performance under dynamic clinical conditions has become a critical research priority. Recent advances in AI-driven clinical decision support have emphasized the importance of uncertainty-aware learning, confidence estimation, and adaptive decision-making to improve medication selection and personalized treatment recommendations (Eguia et al., 2024). Similar concepts are observed in observer-based estimation and adaptive control frameworks, where system states are continuously monitored and updated to maintain reliable performance under disturbances, incomplete information, or changing operating conditions. These principles are highly relevant to healthcare NLP, where models frequently encounter noisy clinical narratives, missing data, domain shifts, and evolving medical knowledge (Lambert et al., 2024). Emerging approaches such as uncertainty quantification, confidence calibration, continual learning, retrieval-augmented generation, and human-in-the-loop validation offer promising mechanisms for improving the robustness of clinical NLP systems. Future research should focus on integrating adaptive learning and uncertainty-aware decision support strategies to enhance model reliability, safety, and clinical applicability in real-world healthcare environments (Barua, Balaji and Balaji, 2026).

A study highlighted the value of uncertainty-aware NLP in clinical practice. In medication safety, NLP systems extract medication histories and allergy information from free-text clinical notes and provide confidence scores for each prediction; low-confidence outputs are flagged for clinician review, reducing medication-error risk and improving concordance with expert assessment. In clinical triage, NLP models identify symptoms and high-risk presentations from unstructured records, and calibrated confidence estimates enable uncertain cases to be escalated for human evaluation improving sensitivity for high-risk patients while minimizing unnecessary alerts (Malarvizhi, Smith and Yang, 2026). In psychiatry, NLP applied at scale to electronic health records identified negative symptoms in schizophrenia and associates them with hospitalization and readmission outcomes, demonstrating how unstructured text can yield actionable clinical insights. Collectively, these examples show that uncertainty quantification, confidence calibration, and human-in-the-loop validation improve safety, reliability, and clinical utility of NLP in real-world healthcare, supporting adaptive, uncertainty-aware decision support (Vaccaro et al., 2026).

10.10. Hallucination and Reliability Risks

One of the key limitations of generative models is that they generate fluent and unsupported or false output, otherwise known as hallucinations (Özer, 2024). Hallucinations are very dangerous in the medical field, where they can present falsified clinical information, false causation, and misleading summaries that can seem reasonable to the user. This has been a point of concern in decision support, clinical summarization and patient-facing applications (Joseph, Jose and Jose, 2025). Contrary to the classic NLP systems that retrieve the information within text, LLMs can generate content outside of the presented evidence unless it is specifically limited. To reduce the risk of hallucination, grounding mechanisms, confidence signaling, and heavy human-in-the-loop supervision are necessary, in particular in a high-stakes application (Bélisle-Pipon, 2024).

Figure 11 is divided into two parts: a pie chart on the left and a bar chart on the right. The pie chart depicts the proportional usage of various knowledge integration methods in biomedical NLP research. Ontologies dominate this landscape, accounting for 55.9% of usage, underscoring their central role in structuring and standardizing biomedical concepts. Other commonly employed methods include customized dictionaries and pretrained embeddings, each contributing 14.7%, which aid in enhancing model vocabulary and contextual understanding. Less frequently used yet significant techniques include ML-BERT-based models (5.9%) and a variety of miscellaneous approaches (8.8%), indicating a trend towards integrating domain-specific language models and hybrid methods.

The bar chart on the right side of the figure complements this analysis by detailing the frequency of ontology usage across different biomedical knowledge sources. Among these, the Unified Medical Language System (UMLS) emerges as the most widely utilized ontology, reflecting its broad coverage and integration of multiple biomedical vocabularies. It is followed by other prominent sources like SNOMED-CT, Medical Subject Headings (MeSH), Chemical Entities of Biological Interest (ChEBI), and UniProt, each serving distinct roles in encoding diseases, medical terms, chemicals, and protein data, respectively. The frequency is based on the number of supporting research papers, demonstrating the strong empirical foundation underpinning ontology-based knowledge integration. Altogether, the figure underscores the essential function of structured domain knowledge in enhancing the accuracy, interpretability, and performance of NLP models in biomedical applications.

11. Current and Future Progress of NLP

Addressing these limitations not only informs ethical implementation but also points toward future research directions that can strengthen the clinical translation of NLP technologies. Current research in NLP focuses on various aspects, such as syntactic phenomena that deal with sentence structure and word order based on grammatical classes rather than meaning. This includes the development of discriminative models for scoring passes, efficient approximate parsing techniques, and dependency grammar. Another area of research is machine translation, which involves developing models and algorithms for translating between languages, particularly for low-resource and morphologically complex languages. Semantic phenomena refer to aspects of language that are relevant to the meaning of a sentence, regardless of the surrounding context. Healthcare tools are shown in Table 6.

Figure 12 provides a holistic visualization of the transformative roles that NLP plays across six pivotal domains in healthcare: precision medicine, patient experience, clinical decision support, medical information, diagnostic support, and research acceleration. Each of these domains harness NLP's capacity to process and analyze unstructured textual data, thereby enabling a range of targeted healthcare applications. In precision medicine, NLP tools facilitate the design of personalized treatment plans by extracting and interpreting genomic data, clinical notes, and patient-specific variables. The patient experience is enhanced through sentiment analysis and patient feedback mining, which helps healthcare providers align services with patient needs and perceptions. The figure also emphasizes the diverse array of state-of-the-art NLP tools and language models employed across these domains, such as BERT, GPT-3, cBERT, Roberta, Transformers, MedBERT, BioGPT, ClinicalT5, BERT-Medical, ClinicalBERT, SciBERT, AllenNLP, and IBM Watson. These models represent the cutting edge of biomedical NLP, enabling accurate interpretation, reasoning, and language understanding within clinical and research settings. Collectively, the figure

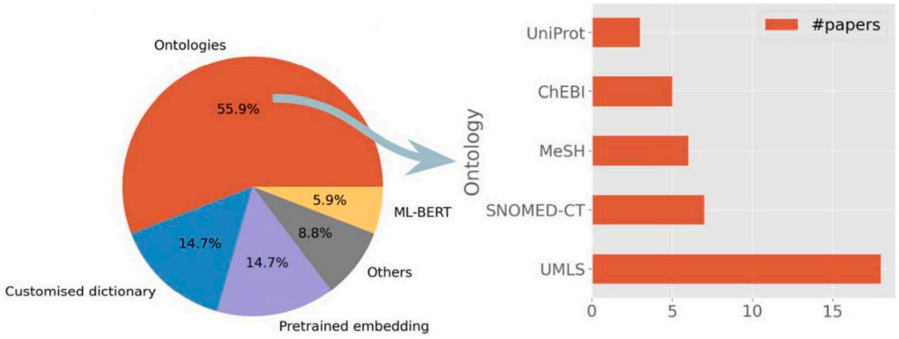


Figure 11. The figure highlights the predominance of ontology-based approaches in biomedical text mining, with ontologies accounting for 55.9% of knowledge utilization techniques, followed by customized dictionaries and pretrained embeddings at 14.7% each, other methods at 8.8%, and ML-BERT models at 5.9%. Among the ontologies applied, UMLS is the most widely used, featuring in nearly 18 studies, while SNOMED-CT and MeSH appear in around 7 and 6 studies, respectively. More specialized ontologies (such as ChEBI and UniProt) are less common, with only 4 and 2 studies utilizing them. Overall, the data suggest that ontologies, particularly UMLS, remain the dominant resource in biomedical NLP, reflecting their comprehensive coverage and strong integration into text-mining workflows (Wu et al., 2022).

Table 6
Tools used in NLP in the healthcare system.

Sr. No.	Tools	Applications	Language Understanding Method	Healthcare Benefit	Reference
1	BERT	Precision medicine, Diagnostic support	Bidirectional encoder represent-ations	Personalized therapy, improved diagnostics	(Babu and Boddu, 2024)
2	GPT-3	Precision medicine, Text generation	Generative pre-training	Personalized treatment plans, automation	(Baig et al., 2024)
3	cBERT	Precision medicine	Domain-specific language modeling	Genetic risk assessment, targeted therapy	(Luo, Deng, Yang and Luo, 2024)
4	Transformers	Patient experience (sentiment analysis, communic-ation)	Contextual embeddings	Improved patient satisfaction and communication	(Perikos and Diamantopoulos, 2024)
5	RoBERTa	Patient experience	Contextual language understanding	Better sentiment detection	(Gardazi et al., 2025)
6	IBM Watson	Clinical decision support	Hybrid NLP & rule-based methods	Real-time diagnostics, treatment recommend-ations	(Thamma, 2025)
7	MedBERT	Clinical decision support	Medical context pre-training	Risk stratification, diagnostic assistance	(Levra et al., 2025)
8	BioGPT	Medical information (scientific writing)	Biomedical text generation	Writing support, knowledge summariz-ation	(Ramoni et al., 2024)
9	ClinicalT5	Medical note summarization	Text-to-text transfer	Efficient document-tation and summariz-ation	(Van Veen et al., 2023)
10	BERT-Medical	Diagnostic support	Medical concept understanding	Accurate medical coding, improved risk prediction	(Li et al., 2023)
11	ClinicalBERT	Diagnostic support	Clinical note contextual modeling	Enhanced diagnosis and patient care	(Talebi et al., 2024)
12	SciBERT	Research acceleration	Scientific literature language modeling	Improved literature analysis and knowledge discovery	(Kumar et al., 2024)
13	AllenNLP	Research acceleration	Modular language understanding tools	Clinical trial matching, fast NLP model prototyping	(Gardner et al., 2018)

illustrates how the strategic integration of NLP technologies is reshaping modern healthcare by augmenting both operational efficiency and clinical intelligence.

12. Summary

Future advancements in NLP will involve deeper integration with mathematical models, algorithm development, and real-world data testing. Key enhancements include expanding backend functionalities such as information extraction, normalization, and database construction, alongside augmenting interface usability. Portable devices with real-time translation and subject search capabilities are expected to support personalized patient interactions. Predictive analytics will evolve to not only forecast disease outbreaks and readmissions but also predict individual responses to treatments based on historical data. NLP will enhance clinical decision support systems by offering real-time, personalized recommendations and further automating documentation to reduce administrative burdens on clinicians (Yagamurthy and Azmeera, 2023). There will be significant improvements in interactive chatbots and virtual health assistants, capable of monitoring chronic diseases and offering tailored advice. Semantic and interoperable health records will improve data sharing and care coordination across platforms. NLP will also accelerate drug discovery by mining scientific literature and clinical trial data for therapeutic insights (Khude and Shende, 2025). Enhanced translation capabilities will support international collaboration, while emotion recognition and sentiment analysis of social media may enable early detection of mental health issues. Tailored educational materials and ethical compliance tools will improve patient understanding and data governance.

Beyond clinical care, NLP will transform healthcare operations, including supply chain management, vendor negotiations, procurement, and fraud detection. AI-driven bots could simulate negotiation strategies, draft contracts, and monitor communications for compliance. Finally, NLP will deliver real-time, context-aware translation, communication analysis, and global trend monitoring to help healthcare organizations stay ahead in an increasingly complex digital landscape

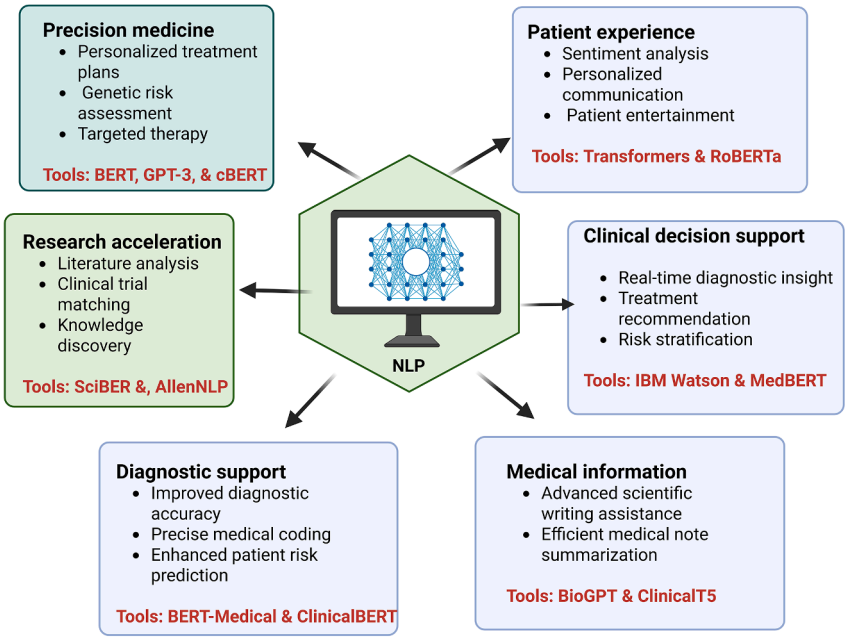


Figure 12. This figure shows how **NLP** supports healthcare across six major areas. In **Precision Medicine (a)**, NLP tools like BERT, GPT-3, and cBERT enable personalized treatment, genetic risk assessment, and targeted therapies. For **Patient Experience (b)**, models such as Transformers and RoBERTa help in sentiment analysis, personalized communication, and patient engagement. In **Clinical Decision Support (c)**, tools like IBM Watson and MedBERT provide real-time diagnostic insights, treatment recommendations, and risk stratification. **Medical Information (d)** is enhanced by BioGPT and ClinicalT5, which assist in scientific writing and summarizing medical notes. **Diagnostic Support (e)** benefits from BERT-Medical and ClinicalBERT to improve diagnostic accuracy, medical coding, and patient risk prediction. Finally, **Research Acceleration (f)** leverages SciBERT and AllenNLP for literature analysis, clinical trial matching, and knowledge discovery. Together, these applications highlight how NLP drives better patient care, clinical efficiency, and medical research (Eguia et al., 2024) .

(Olorunniwo, Alonge and Isreal, 2025).

13. Conclusion

NLP has emerged as a transformative technology in healthcare, enabling advances across these domains. The growing integration of NLP into clinical workflows demonstrates its potential to improve information accessibility, support evidence-based decision-making, and enhance healthcare delivery. Nevertheless, widespread adoption remains constrained by challenges related to data quality, interoperability, fairness, explainability, privacy, workflow integration, and validation in real-world clinical settings.

Clinical NLP will require coordinated efforts across multiple stakeholder groups. Researchers should prioritize robust external validation, explainable and uncertainty-aware models, multilingual healthcare applications, and standardized evaluation frameworks that assess clinical utility alongside technical performance. Clinicians and healthcare organizations should adopt human-in-the-loop deployment strategies, invest in AI literacy and workforce training, and implement continuous monitoring mechanisms to ensure safe and effective use in practice. Policymakers and regulators should establish harmonized governance frameworks addressing transparency, auditability, fairness, privacy, interoperability, and post-deployment surveillance. In the near term, prioritizing human-in-the-loop validation and uncertainty-aware models will be critical for safe deployment; long-term success depends on multilingual corpus development and harmonized regulatory frameworks. Through such collaborative efforts, NLP can evolve from promising research innovations to trustworthy, equitable, and scalable healthcare solutions that improve patient care and clinical outcomes across diverse healthcare environments.

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Author Contributions

DK, VG, KB, RT: writing-original draft preparation; NC, SA, TJW: writing-review and editing; SJ, MSS, SG: Data curation, formal analysis; MF, TJW: Conceptualization, investigation, supervision, project administration. All authors have read and agreed to the published version of the manuscript

Abbreviations

The following abbreviations are used in this manuscript:
Machine Learning (ML)
Natural Language Processing (NLP)
Deep Learning (DL)
Optical Character Recognition (OCR)
Named Entity Recognition (NER)
Electronic Health Records (EHR)
Term Frequency–Inverse Document Frequency (TF-IDF)
Linguistic Inquiry and Word Count (LIWC)
Area Under the Curve (AUC)
Support Vector Machine (SVM)
Convolutional Neural Network (CNN)
Computed Tomography (CT)
Magnetic Resonance Imaging (MRI)

Declaration of competing interest

The authors decleare no conflict of interest.

Data availability

No datasets were generated or analyzed during the current study. All information discussed in this article is derived from previously published sources, which are properly cited.

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